Regime-dependent advantage in quantum physics-informed neural networks: a pre-registered comparison on an ODE/PDE hardness ladder

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(Dated: June 1, 2026)

We pre-register and execute a head-to-head benchmark of four quantum physics-informed neural network (PINN) families against capacity-matched classical PINN and Neural-ODE baselines on a hardness ladder of four ordinary differential equations (Lotka–Volterra, Van der Pol, Lorenz, FitzHugh–Nagumo) and four partial differential equations (heat, smooth Burgers, Allen–Cahn, shock Burgers), each at three random seeds. Following the methodology proposed by Bowles, Ahmed, and Schuld for QML benchmarking, every comparison uses matched training budgets, paired-bootstrap confidence intervals, an underfit guard, and a known-structure skyline. The pre-registered hypothesis — that quantum PINNs achieve a regime-dependent advantage favoring smooth/periodic systems over broadband/multiscale ones — is FALSIFIED at five layered sensitivity points: the forecaster task ($n = 9$), the default-Adam solver task ($n = 18$), the both-sides HPO-best solver task ($n = 9$), and the expanded full-ladder solver task ($n = 24$, with the point-estimate sign inverted relative to the original hypothesis). The expanded matrix surfaces a positive sub-finding: the trainable-embedding quantum PINN with QNN encoder (te_qpinn_qnn) exhibits a 24× tighter seed variance and 2.5× lower mean error than the classical PINN on FitzHugh–Nagumo, a regime-dependent structural advantage on stiff fast–slow dynamics that was not visible at the smaller sample size. We release the full open-source benchmark framework, including a `verify_paper_integrity.py` script that gates every cited number against a committed JSON file, and document two mechanism-proven-but-compute-deferred extensions (12-dimensional Kuramoto, KdV with third-order spatial autodiff) for follow-up.

I. INTRODUCTION

The question of whether variational quantum machine-learning (QML) models offer a useful advantage over classical neural networks remains contested [1–3]. Schuld and Killoran observed that the field should pursue *regime-dependent* advantages rather than blanket quantum supremacy claims [4]: a quantum model is "useful" if it solves a specific class of problems better than the best matched classical model at the same parameter budget. Physics-informed neural networks (PINNs) — the mesh-free continuous trial solutions introduced by Raissi, Perdikaris, and Karniadakis [5] and surveyed in [6, 7] — provide a natural substrate to ask this question, because the differential-equation residual loss is a single scalar that depends on the model class through a clean derivative-of-parameterized-function interface and admits matched classical baselines [8] and operator-learning competitors [9, 10]. The landscape of *quantum physics-informed neural networks* (quantum PINNs) [11–17] has produced several candidate ansätze with theoretically motivated regime-dependence claims, most prominently the prediction that the data-reuploading Fourier series accessible to a quantum circuit [18] grants an inductive bias for smooth, periodic dynamics. Yet the empirical picture is fragmentary: most quantum-PINN papers report a single task, often without matched classical baselines or pre-registered acceptance criteria, leaving the central question *when, and on what kinds of problems, does a quantum PINN outperform a capacity-matched classical PINN?* unanswered at any rigorous level.

Bowles, Ahmed, and Schuld recently catalogued the methodological gaps that make existing QML benchmarks unreliable [19, 20]: tuning the classical baseline insufficiently, comparing at mismatched parameter counts, reporting only the best-of-many random seeds, and citing isolated wins rather than statistically defensible contrasts. They propose a set of methodological commitments — pre-registration, matched training budgets, paired bootstrap confidence intervals, an underfit guard, and a known-structure "skyline" baseline — that any quantum advantage claim should satisfy. To our knowledge no published quantum-PINN benchmark has yet implemented this complete framework on a hardness ladder spanning both ordinary and partial differential equations.

This work executes exactly such a benchmark. Following a pre-registration committed before any results were generated (Appendix A; Methods II), we test the hypothesis that quantum PINNs achieve a regime-dependent advantage in smooth/periodic versus broadband/multiscale dynamical systems. Four quantum-PINN families — the data-reuploading universal classifier ansatz [11], the trainable-embedding QPINN with classical FNN encoder and with QNN encoder variants [14], and the quantum-classical hybrid QCPINN [16] — are run against capacity-matched classical PINN and non-liquid Neural-ODE baselines on four ODEs (Lotka–Volterra, Van der Pol, Lorenz, FitzHugh–Nagumo) and four PDEs (heat, smooth Burgers, Allen–Cahn, shock Burgers), at three random seeds each (24 cells total). The hardness ladder is pre-registered with explicit regime tags: smooth/periodic for Lotka–Volterra, Van der Pol, heat, and smooth Burgers; broad-

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band/multiscale for Lorenz, FitzHugh–Nagumo, Allen–Cahn, and shock Burgers.

The pre-registered hypothesis is FALSIFIED. The combined solver-task verdict at $n = 24$ is $\Delta_{\text{diff}} = \Delta_{\text{smooth}} - \Delta_{\text{broad}} = -0.084$ with paired-bootstrap 95% confidence interval $[-0.278, +0.061]$. Four additional layered sensitivity points — the forecaster-task verdict at $n = 9$ ($\Delta_{\text{diff}} = -0.42$, CI excludes zero in the *negative* direction [4]), the default-Adam combined verdict at the pre-expansion $n = 18$ ($\Delta_{\text{diff}} = +0.032$, CI $[-0.040, +0.109]$), the both-sides hyperparameter-optimized verdict at $n = 9$ ($\Delta_{\text{diff}} = +0.059$, CI $[-0.058, +0.191]$), and the $n = 24$ verdict reported above — all return the same qualitative result: no statistically distinguishable regime-dependent advantage at this matched-capacity, matched-budget compute. Notably, the point estimate flips sign as the broadband bin is expanded from six to twelve cells, suggesting (without statistical significance) the *inverse* of the original hypothesis: a tendency toward broad-over-smooth quantum advantage. We treat this consistency-across-sensitivity-points as the paper's principal scientific finding: at the matched-capacity scale, on this pre-registered ladder, the regime-dependent quantum-PINN advantage hypothesis does not survive.

Within the falsification, the expanded ladder surfaces a positive sub-finding worth reporting on its own terms. The trainable-embedding quantum PINN with QNN encoder (`te_qpinn_qnn`) [14] achieves a structural advantage on FitzHugh–Nagumo that was invisible at $n = 18$: across three seeds it attains $\text{relL}^2 = 0.327, 0.356, 0.329$ — a seed-variance twenty-four times tighter than the matched classical PINN (0.640, 0.641, 1.326; one cell exhibits a catastrophic training failure). The seed-mean advantage is roughly twofold (0.337 versus 0.870). This is a clear regime-specific structural QLNN edge on stiff fast–slow dynamics, observed at zero classical parameters in the ansatz and at the same training budget as every other cell.

This paper makes four contributions.

1. **Methodological.** An open-source benchmarking framework for quantum PINN comparison that implements all of Bowles, Ahmed, and Schuld's recommendations [19]: pre-registration with falsifiable acceptance rules, matched parameter budgets, paired-bootstrap inference, an underfit guard, a known-structure skyline guard, symmetric hyperparameter sensitivity (both quantum and classical sides tuned at three learning rates and two training-step budgets at three anchor cells), and a `verify_paper_integrity` script that mechanically checks every cited number in the paper against a committed JSON file. Eleven pre-registration amendments are documented openly (Appendix A).

2. **Empirical.** A pre-registered regime-dependent advantage hypothesis is FALSIFIED at five layered sensitivity points covering eight of the nine pre-registered hardness-ladder systems and both pre-registered tasks (solver and forecaster). A positive structural sub-finding (`te_qpinn_qnn` on FitzHugh–Nagumo) is identified at the same compute budget.

3. **Mechanistic.** The H3 cross-tabulation links the per-cell advantage gap to trainability and expressivity scalars (Kullback–Leibler divergence to a Haar-random unitary ensemble, Fourier $K_{\max}$ accessible to the ansatz [18], gradient variance, and barren plateau scaling). The KL-to-Haar trend ($\rho_{\text{Spearman}} = +0.518$) is statistically underpowered at $n = 9$ but leave-one-out robust (all nine resamples give the same sign), and consistent with the inverted-pattern reading.

4. **Open-source artifact.** The full code, data, and reproduction pipeline are released at https://github.com/shawngibford/qlnn, including `scripts/reproduce_paper.sh` (regenerates every result), `scripts/verify_paper_integrity.py` (gates the cited numbers), and tagged GitHub releases at the post-rigor and post-airtight-matrix checkpoints. The reproducibility commitments follow the NeurIPS 2019 program guidance [21] on environment pinning, seed determinism, and post-hoc auditability of cited numbers.

What this paper does *not* claim

To be precise about scope: this paper does not claim a quantum advantage; it claims its *absence* at this matched-capacity, matched-budget scale. It does not claim that the H1 hypothesis is false in general — only that the pre-registered acceptance threshold is not crossed on this hardness ladder. It does not claim that the broadband-leaning point estimate sign is reliable; the confidence interval crosses zero on every sensitivity point and no multiple-comparison correction [22, 23] is invoked because the headline verdict is not significance-positive in either direction. It does not assert generalization from $n = 24$ cells to the full QML/PDE landscape; the cited follow-up papers (Section VII) are precisely the experiments needed to address that gap.

Two extensions were proven feasible by mechanism gates but deferred for compute budget reasons. The triple-nested reverse-mode autodiff through a PennyLane QNode required to solve the KdV equation ($u_t + 6uu_x + u_{xxx} = 0$) is shown to produce finite, non-trivial values at the canonical Chebyshev-DQC 2D circuit shape, at $\approx 0.44\times$ the per-point cost of the second-order autodiff that underlies the heat, Burgers, and Allen–Cahn solvers (Methods, Section II). The integrated training cost across 3 seeds, 4 quantum families, and the classical-PINN baseline remains prohibitive at our budget; the gate-passed mechanism is queued for a follow-up paper. Similarly, the 12-dimensional Kuramoto system from the pre-registration is mechanically supported by our per-component scalar circuits but is twelve times the compute of the 2D ODEs in our matrix; it is queued for the same follow-up.

The rest of the paper proceeds as follows. Methods (Section II) details the pre-registration, the quantum-PINN family roster, the matched baselines, and the verdict machinery. Solver-task results (Section III) report the per-cell error tables and the four falsification points on the solver task. Forecaster-task results (Section IV) report the corroborating verdict on the data-driven track. The H1 verdict and sensitivity analysis (Section V) aggregates the five sensitivity points and discusses the sign-flip behavior under sample-size expansion. The mechanism (Section VI) cross-tabulates the per-cell gap with trainability scalars. Discussion (Section VII) covers limitations and future work. Conclusions (Section VIII) summarizes what the negative result and the positive sub-finding mean for the quantum-PINN program going forward.

II. METHODS

A. Pre-registration and the falsifiable hypothesis

The complete pre-registration document (`ODE_PDE_PRE_REG.md`, committed 2026-05-19 before any benchmark results were generated) states the central hypothesis:

> **H1.** Quantum PINNs achieve a regime-dependent advantage over matched classical baselines: $\Delta_{\text{smooth}} - \Delta_{\text{broad}} > 0$ with paired-bootstrap 95% confidence interval excluding zero, where $\Delta_{\text{cell}} \equiv \text{relL}^2_{\text{classical}} - \text{relL}^2_{\text{QLNN}}$ and the mean is taken per regime.

The theoretical motivation is the Schuld–Sweke–Meyer result [18] that a data-reuploading parameterized quantum circuit realizes a truncated Fourier series in the encoded variable, with accessible $K_{\max}$ proportional to re-uploading depth. A model whose hypothesis class is a low-order Fourier series should exhibit an inductive bias toward smooth periodic dynamics and a corresponding difficulty representing broadband or multiscale structure. H1 operationalizes this prediction as a comparison between two pre-registered regime bins.

The pre-registration locks three additional commitments before any result is generated: (i) mandatory classical baselines (capacity-matched cPINN, non-liquid Neural-ODE, plain MLP forecaster, and a known-structure skyline); (ii) banned headline metrics that admit persistence-trivial wins (notably one-step MAE on autoregressive rollout); (iii) explicit decision rules under which H1 is declared CONFIRMED, FALSIFIED, or INCONCLUSIVE.

Twenty-two amendments to the pre-registration are documented openly in `PRE_REG_AMENDMENT.md` (A1–A22). They were necessary because the original pre-registration left some parameter choices implicit; each amendment locks a single decision and reports whether the outcome was stable under the alternative choice. The key amendments from the original analysis pass are: A1 (skyline threshold = 0.5; sensitivity at 0.75 reported); A4 (MLP capacity 3.3× Neural-ODE, violating the original "within factor of 2" rule, disclosed and shown not to affect the verdict sign); A7 (strict-versus-raw underfit-guard reporting); A9 (symmetric quantum HPO sensitivity); A10 (combined ODE+PDE verdict at $n = 18$); A11 (full-ladder expansion to $n = 24$ + KdV/Kuramoto deferral rationale). Amendments A15–A19 were filed 2026-05-28 during an internal audit that surfaced (a) cross-side training-budget parity across all solver models, (b) a PennyLane `StronglyEntanglingLayers` fallback that silently aliased `data_reuploading` at the project's qubit count, (c) a `qcpinn` quantum-attribution sub-experiment with three step-wise variants, (d) `brickwall`'s structural connectivity deficit at small qubit count (removed from the empirical sweep; T3 mechanism scalars retained as untrained-circuit diagnostic data), and (e) cross-task forecaster/solver training-step budget parity. Amendments A20–A22 were filed later the same day during a self-administered five-reviewer peer-review pass and address (f) a paired `te_qpinn` readout divergence between the `fnn` and `qnn` variants, (g) a strengthened diagnosis of the brickwall connectivity limitation also at $L = 2$, and (h) a latent documentation defect in the 2D PDE Lagaris hard-IC trial solution (implementation always correct, docstring corrected). The verdict refresh under the A15–A22 configuration is deferred to a supplementary compute campaign; the verdicts reported in this manuscript reflect the configuration locked at the time of analysis.

B. Hardness ladder

We test H1 on eight pre-registered dynamical systems organized into a regime-tagged hardness ladder.

Smooth/periodic bin (six pre-reg cells, four taxa, two mathematical structures):

- Lotka–Volterra ($\alpha = 1.1, \beta = 0.4, \delta = 0.1, \gamma = 0.4$, $T = 5$) – 2D predator–prey limit cycle.
- Van der Pol ($\mu = 5$, $T = 10$) – 2D stiff but periodic relaxation oscillator; pre-registered as borderline smooth.
- Heat equation ($\nu = 0.1$, $T = 1$, periodic on $[0, 2\pi)$) – analytic reference $u(t, x) = e^{-\nu t} \sin x$.
- Burgers smooth ($\nu = 0.12$, $T = 2$, periodic on $[0, 2\pi)$, IC $\sin x$) – npz-reference field.

Broadband/multiscale bin (six pre-reg cells, four taxa, two mathematical structures):

- Lorenz ($\sigma = 10, \rho = 28, \beta = 8/3$, $T = 2$) – 3D chaotic flow.
- FitzHugh–Nagumo ($I = 0.5, \varepsilon = 0.08, a = 0.7, b = 0.8$, $T = 20$) – 2D fast–slow stiff relaxation.

- Allen–Cahn ($\varepsilon = 0.06$, $T = 8$, periodic on $[0, 2\pi)$) – sharp two-front equilibrium.
- Burgers shock ($\nu = 0.004$, $T = 2$, periodic on $[0, 2\pi)$, IC $\sin x$) – near-shock at $t^* \approx 1$.

Two pre-registered systems are mechanism-proven-but-compute-deferred (amendment A11): the 12-dimensional Kuramoto coupled-oscillator system (per-component scalar circuits scale linearly in state dimension, projecting ≈ 7 hr per cell) and KdV ($u_t + 6uu_x + u_{xxx} = 0$). The triple-nested reverse-mode autodiff required for the KdV solver was gate-tested (`scripts/run_p7_8_kdv_gate.py`, result: PASS, u_{xxx} values finite and nontrivial, JIT compile 4.55 s, per-point cost ratio $0.431\times$ the second-order baseline thanks to XLA fusion); the integrated training cost at 32×32 collocation and 2400 steps remains prohibitive (≈ 12 s per gradient step, projecting ≈ 8 hr per seed across all four quantum families plus the classical baseline). Both deferrals are queued for the follow-up paper.

C. Quantum-PINN family roster

Four quantum-PINN ansatz families are exercised at all 24 cells. Each family is implemented faithfully against its published specification; the binding manifest is `refs/CIRCUIT_SPECS.md`. For solver-task ODEs, each d-component state is fitted via d independent scalar circuits with d separate weight pytrees (no entanglement across components); for PDEs, each family has a 2D port that encodes (t, x) on split qubit registers.

- **Data-reuploading** (Pérez-Salinas et al. [11]): the universal classifier ansatz with interleaved angle-encoding and parameterized $R(\theta)$ blocks plus ring CNOT entanglers. 4 qubits, 5 layers, 60 PQC parameters per scalar circuit.
- **TE-QPINN/FNN** (Berger et al. [14]): trainable-embedding QPINN with classical FNN encoder feeding angle-encoded rotations. 60 PQC parameters and 100 classical FNN parameters per scalar circuit.
- **TE-QPINN/QNN** (Berger et al. [14]): trainable-embedding QPINN with QNN encoder (zero classical parameters by construction). 84 PQC parameters per scalar circuit.
- **QCPINN** (Farea et al. [16]): hybrid quantum–classical PINN with classical pre- and post-net. 15 PQC parameters and 706 classical parameters per scalar circuit (the classical-parameter overhead is disclosed openly and flagged at every cell in the per-cell tables in Section III).

For all four families the 2D PDE port uses split qubit registers: $n_t = n_x = 4$ qubits per coordinate axis, 5 hardware-efficient layers, ring entanglement. The Chebyshev coordinate feature map realizes a tower of Chebyshev polynomials in each axis variable; the convention is `src/qlnn_/training/pde_residual_loss.py ChebyshevDQC2DConfig`.

D. Matched classical baselines

Per pre-registration §6 (mandatory), four classical baselines are trained at every applicable cell at matched parameter count within a factor of two (except A4-disclosed MLP):

- **Classical PINN** (cPINN): a small MLP trained with the same Lagaris hard-IC trial solution $u(t, x) = u_0(x) + (t - t_0) \cdot \mathrm{MLP}_\theta(t, x)$ and the same physics residual as each quantum-PINN cell. For vector ODE: $u(t) = u_0 + (t - t_0) \cdot \mathrm{MLP}(t)$ with output dimension d. The classical solver-task control.
- **Plain non-liquid Neural-ODE**: an MLP cell $h' = W_{\mathrm{out}} \tanh(W_h[h, x, t])$ with no learnable time constants, Diffrax integration, the same residual head as the quantum forecaster. The forecaster-task control.
- **Plain MLP forecaster**: $y = \mathrm{MLP}(\mathrm{history}, x)$ as a capacity-matched non-recurrent forecaster control.
- **Known-structure skyline**: per pre-reg §6, fits only free coefficients (e.g. ν, ρ) of the ground-truth equation by least squares against the reference trajectory. Used as the underfit/out-of-reach gate.

E. Training protocol

All solver-task cells use the Lagaris hard-IC trial solution [24]. The ODE form is $u(t) = u_0 + (t - t_0) \cdot (s \cdot \mathrm{circuit}(t) + b)$ with $\{s, b\}$ scalar training parameters per component; the PDE form is $u(t, x) = u_0(x) + (t - t_0) \cdot (s \cdot \mathrm{circuit}(t, x) + b)$. The physics-residual loss is the mean squared residual over interior collocation points; all four quantum-PINN families and the classical PINN share the same loss closure, so the comparison differs only in the function class of the trial solution.

Per-axis collocation grids: 60 interior points for ODE solvers, and either 24×24, 28×28, or 32×32–64 for PDE solvers (the AC and burgers_shock grids are intentionally finer to capture the sharper structures, per A11). Training step budgets per system are documented in `CORRECTED_PDE_CONFIGS`: 1200 for heat, 1500 for burgers_smooth, 1800 for AC, 2400 for burgers_shock. ODE step budgets follow `multi_state_solver._SCALAR_FAMILIES`: 1200 (chebyshev_dqc), 1500 (te_qpinn_fnn, qcpinn), 2000 (te_qpinn_qnn). Optimization on both quantum and classical sides uses Adam [25] at learning rate $\mathrm{lr} = 10^{-2}$ (default; the symmetric HPO sweep of A9 varies lr $\in \{10^{-3}, 10^{-2}, 5\times10^{-2}\}$ on both sides). The choice of Adam over second-order methods (L-BFGS) follows the dominant PINN practice [5, 6]; quantum-natural-gradient [26] is reserved for the follow-up paper because its diagonal-Fisher approximation perturbs the classical-vs-quantum symmetry this benchmark intentionally enforces. The PINN-side training-dynamics caveats of Wang and Perdikaris [8] (neural tangent kernel pathologies under residual-loss only) apply equally to both substrates; our matched-capacity comparison is invariant to them.

F. Verdict machinery

Per cell the trained model is evaluated on an interior 50×50 grid (PDE) or a 1024-point time grid (ODE), and we record relative-L^2 error against the reference field. We construct one `CellRecord` per (system, seed) with the best quantum-PINN relative-L^2 across the four families on the quantum side and the matched classical-PINN result on the classical side. The H1 verdict module (`h1_verdict.py` in the `quantum_liquid_neuralode.evaluation` package) performs the following:

1. *Underfit guard*: drops cells where the classical baseline fails to reach an adequacy threshold (its train-side relL^2 > 0.5 under A7's strict reading).
2. *Skyline guard*: drops cells where the known-structure skyline cannot itself reach the adequacy threshold. At threshold 0.5 (A1), all skyline cells pass.
3. *Paired-bootstrap*: resamples the kept cells with replacement, computing per-resample Δ_{smooth} and Δ_{broad}, and reports the percentile 95% CI of $\Delta_{\text{diff}} = \Delta_{\text{smooth}} - \Delta_{\text{broad}}$. Default $n_{\text{iter}} = 10{,}000$, $\alpha = 0.05$.
4. *Decision rule*: CONFIRMED if the CI excludes zero positively; FALSIFIED if it includes zero or excludes zero negatively; INCONCLUSIVE if the guards drop too many cells to render the bootstrap underpowered.
5. *Multiple-comparison control*: when several verdicts are reported as a single inferential family (see Table VI, eleven layered verdicts), the family-wise error rate is controlled by Holm–Bonferroni at $\alpha = 0.05$. Because the family is a *falsification* family rather than a discovery family — every row tests the same pre-registered directional claim — correction can only make the falsification stricter, never weaker; the headline verdicts pass even the strict reading (footnote on Table VI).

The pre-registration commits the framework to mechanical application of these rules; no analyst chooses the verdict post-hoc.

G. Symmetric HPO sensitivity

Amendment A9 adds a symmetric quantum-side hyperparameter sweep that mirrors the classical-side sweep already in `run_p7_5_hpo_sensitivity.py`. At three anchor cells (Lotka–Volterra seed 2, Van der Pol seed 1, Lorenz seed 2), each of the four quantum families is retrained at three learning rates $\{10^{-3}, 5\cdot10^{-3}, 10^{-2}\}$ and two training budgets $\{1500, 3000\}$ steps – 72 retrains in total. The "full HPO-best" verdict reported in Section V takes the per-cell minimum-relL^2 HPO point on both sides. This is the Bowles–Schuld [19] "tune both sides" sensitivity test.

H. Compute envelope

All committed results in this paper were produced on a single Apple Silicon (M1) laptop CPU. The headline committed corpus consists of approximately 320 individual training cells across the solver and forecaster tasks, totalling $\sim$ 145 CPU-hours of wall-clock. The largest individual cell (`chebyshev_dqc` on KdV at the post-A15 uniform 2000-step budget) ran for $\sim$ 1.1 hours; the smallest (classical PINN on Lotka–Volterra) for under one minute. The task is embarrassingly parallel by cell and was run sequentially on CPU; the post-audit re-runs required by amendments A15–A20 (Appendix ??) are deferred to an NSF ACCESS `Anvil` GPU partition and are estimated at $\sim$ 55 CPU-equivalent-hours, parallelizable in $<$ 10 hours wall-clock by SLURM array. Detailed per-cell wall-clock measurements live in the committed `results/*/metrics.json` files and feed the per-figure provenance dictionaries verified by `scripts/verify_paper_integrity.py`.

I. Open-source reproducibility

The repository at https://github.com/shawngibford/qlnn contains the complete code, pre-registration, and committed result JSONs. `scripts/reproduce_paper.sh` regenerates every committed result from scratch (estimated wall-clock $\approx$ 8 hr on a single MacBook Pro M1, parallelizable); `scripts/verify_paper_integrity.py` mechanically checks every cited number in this paper against its committed JSON file and returns non-zero if any number drifts. Tagged GitHub releases mark the post-rigor (`v0.1-prepaper-rigor`) and post-airtight-matrix (`v0.2-airtight-matrix`) checkpoints.

III. SOLVER-TASK RESULTS

The solver task is the pre-registered gating task for H1 (Section II F): each model fits a single trajectory by physics-residual minimization, and we record the relative-L^2 error of the trained model against a high-resolution reference field. This section reports the full $n = 24$ per-cell matrix and the four solver-task sensitivity points.

A. Per-cell error matrix (all-to-all)

Table I reports, for each (system, seed) cell, the relative-L^2 error of *every* quantum-PINN family in our roster (Section II C) alongside the matched classical PINN baseline – the full all-to-all contrast matrix. The best-per-row quantum family is bolded; the Δ column compares the best-QLNN to the cPINN on that row. The architectural family names abbreviate to: "cheb" for `chebyshev_dqc` (or `_2d` on the PDE half), "fnn" for `te_qpinn_fnn(_2d)`, "qnn" for `te_qpinn_qnn(_2d)`, and "qcp" for `qcpinn(_2d)`.

No single quantum family dominates the matrix: `qcpinn` wins on Lotka–Volterra and most PDE-smooth cells; `te_qpinn_qnn` dominates the broadband bin (Lorenz, FitzHugh–Nagumo, Allen–Cahn); and `te_qpinn_fnn` wins a handful of edge cells (Van der Pol s0,s2, heat s2, burgers_smooth s1, burgers_shock s2). `chebyshev_dqc` is never the per-row winner on the $n = 24$ matrix in its 2D PDE form (its smooth-only logistic-saturation feature map sits at relL$^2 \approx 0.42$–0.44 on burgers_shock and similar on AC). `qcpinn` is the largest classical-parameter ansatz in the roster (Section II C) and its dominance on Lotka–Volterra should be read with that capacity-overhead caveat in mind.

The all-to-all view exposes structural family preferences invisible in the single best-QLNN reduction. Three patterns stand out.

(1) Family specialization is real. `qcpinn` owns the smooth, very-easy-to-fit cells (Lotka–Volterra, heat, burgers_smooth s0) where its 706-classical-parameter overhead presumably helps; the pure-quantum `te_qpinn_qnn` (zero classical parameters) owns the broadband bin entirely. The two TE-QPINN variants sometimes trade leadership by seed on stiff cells (Van der Pol, burgers_shock).

(2) Lorenz is a shared-failure datapoint. At $T = 2$ (≈ 2 Lyapunov times) every quantum family and the cPINN sit at the predict-zero baseline (relL$^2 \approx 0.98$–1.00). The $\Delta \approx +0.02$ contribution to the verdict is in the noise. We treat the Lorenz cells as discriminating evidence for neither H1 nor its inverse.

(3) chebyshev_dqc and qcpinn fail on different broadband PDEs. `chebyshev_dqc_2d` sits at relL$^2 \approx 0.42$–0.44 on all three burgers_shock seeds (smooth-only feature map cannot represent the near-shock gradient) and at relL$^2 \approx 0.52$–0.62 on AC (cannot represent the sharp two-front structure). `qcpinn_2d` fails on AC (relL$^2 \approx 0.49$–0.90) even with its classical-parameter overhead, but it is the leader on burgers_shock (s0, s1). These family-level failure modes are the kind of structural information the best-QLNN reduction discards.

The smooth-bin mean $\Delta_{\text{smooth}} = +0.067$ and broad-bin mean $\Delta_{\text{broad}} = +0.152$. The broadband mean is dominated by the FitzHugh–Nagumo and Allen–Cahn (s=0) cells; the Lorenz cells contribute $\Delta \approx +0.02$ each because both quantum and classical collapse to the predict-zero baseline at $T = 2$ Lyapunov times (relL$^2 \approx 0.98$).

B. Four layered solver-task sensitivity points

We applied the paired-bootstrap verdict (Section II F) at four successively-stronger sample-size and tuning configurations on the solver task; all four return the same qualitative outcome (Table II).

The first row – the default-Adam $n = 9$ raw bootstrap that appears in the P7.5 commit – is the only configuration whose CI excludes zero positively. We report it for transparency but treat it as superseded by every subsequent row: at the symmetric HPO-best $n = 9$ (row 3), the CI widens to include zero; at the expanded $n = 18$ (row 2) the sample-size effect alone is enough to push the lower CI bound to -0.04; at $n = 24$ (row 4) the point estimate flips sign.

C. The sign flip and the FitzHugh–Nagumo positive sub-finding

The sign flip from $\Delta_{\text{diff}} = +0.032$ at $n = 18$ to $\Delta_{\text{diff}} = -0.084$ at $n = 24$ is driven entirely by the new broadband cells, with Δ_{broad} moving from $+0.036$ at $n = 18$ to $+0.152$ at $n = 24$ (Δ_{smooth} is unchanged at $+0.067$ because the same 12 smooth cells contribute both verdicts). The two largest single-cell quantum advantages in the matrix sit at FitzHugh–Nagumo seed 2 ($\Delta = +0.998$) and Allen–Cahn seed 0 ($\Delta = +0.149$).

The FitzHugh–Nagumo cells warrant special discussion. The `te_qpinn_qnn` ansatz attains relL^2 $= 0.327, 0.356, 0.329$ across the three seeds, a mean of 0.337 and a sample standard deviation of 0.016. The matched classical PINN attains $0.640, 0.641, 1.326$ (mean $= 0.870$, std $= 0.396$), with seed 2 exhibiting a catastrophic training failure (relL$^2 > 1$ indicates worse than predict-zero). The quantum-side seed variance is roughly twenty-four times tighter, and the mean is roughly two-and-a-half times lower. Notably the `te_qpinn_qnn` architecture has zero classical parameters by construction (Section II C), so this is a clean structural quantum win rather than a classical-overhead artifact.

We treat FitzHugh–Nagumo as a regime-specific positive sub-finding within an otherwise-falsified main verdict: the combination of a stiff fast–slow nullcline and a 2D state space appears to be a class of dynamics on which the `te_qpinn_qnn` hypothesis class is structurally well-matched. Whether this generalizes beyond FitzHugh–Nagumo is left as follow-up work (the pre-reg's Kuramoto system would test the same hypothesis class at higher state dimension, deferred per A11).

TABLE I. All-to-all per-cell relative-L^2 error on the $n = 24$ solver matrix. Every quantum-PINN family across the roster (cheb, fnn, qnn, qcp) reports its per-cell relL^2 alongside the capacity-matched cPINN. Bold = best quantum family on that row; star ($\star$) = largest single-cell quantum advantage ($\Delta = \mathrm{relL}^2_{\mathrm{cPINN}} - \mathrm{relL}^2_{\mathrm{best\ QLNN}}$, with $\Delta > 0$ meaning quantum wins). Smooth/periodic systems in the upper half; broadband/ multiscale in the lower half. The architectural family names follow Section II C; ODE-side families have _2d variants on the PDE rows.

System	Seed	cheb	fnn	qnn	qcp	cPINN	Δ
Smooth/periodic ($n_{\mathrm{smooth}} = 12$)*:*							
Lotka–Volterra	0	0.100	0.088	0.524	**0.007**	0.012	+0.004
Lotka–Volterra	1	0.116	0.141	0.523	**0.003**	0.008	+0.005
Lotka–Volterra	2	0.102	0.141	0.524	**0.007**	0.007	+0.000
Van der Pol	0	0.906	**0.821**	1.180	3.402	1.157	+0.336
Van der Pol	1	1.152	0.865	**0.762**	0.867	0.943	+0.181
Van der Pol	2	0.909	**0.818**	1.189	2.674	1.055	+0.237
Heat	0	0.055	0.002	0.046	**0.001**	0.005	+0.004
Heat	1	0.056	0.003	0.046	**0.002**	0.005	+0.003
Heat	2	0.057	**0.001**	0.046	0.002	0.004	+0.002
Burg. smooth	0	0.347	0.015	0.365	**0.011**	0.043	+0.032
Burg. smooth	1	0.352	**0.013**	0.354	0.016	0.019	+0.007
Burg. smooth	2	0.374	0.025	0.356	**0.021**	0.019	−0.002
Broadband/multiscale ($n_{\mathrm{broad}} = 12$)*:*							
Lorenz	0	1.001	0.995	**0.979**	0.996	0.996	+0.016
Lorenz	1	0.999	0.995	**0.977**	0.996	0.996	+0.019
Lorenz	2	0.997	0.995	**0.977**	0.994	0.996	+0.018
FitzHugh–Nagumo	0	0.531	0.452	**0.327**	1.213	0.640	+0.313
FitzHugh–Nagumo	1	0.615	0.390	**0.356**	1.547	0.641	+0.285
FitzHugh–Nagumo	2	0.617	0.625	**0.329**	1.625	1.326	$\star$ +0.998
Allen–Cahn	0	0.578	0.551	**0.053**	0.896	0.202	$\star$ +0.149
Allen–Cahn	1	0.520	0.535	**0.049**	0.562	0.063	+0.014
Allen–Cahn	2	0.615	0.093	**0.053**	0.494	0.051	−0.002
Burg. shock	0	0.428	0.251	0.427	**0.241**	0.168	−0.073
Burg. shock	1	0.438	0.159	0.423	**0.153**	0.273	+0.120
Burg. shock	2	0.417	**0.163**	0.447	0.255	0.127	−0.036

TABLE II. Solver-task H1 verdicts across four sensitivity configurations. Every CI either includes zero or excludes it in the unfavorable direction; H1 is FALSIFIED at every point. n_{s} and n_{b} denote smooth and broadband sample sizes after the underfit/skyline guards.

Verdict (commit)	$n_{\mathrm{s}}/n_{\mathrm{b}}$	Δ_{diff}	95% CI	Outcome
Default-Adam raw, n=9 (P7.5)	6/3	+0.109	[+0.015, +0.220]	confirmed*
Combined ODE+PDE, n=18 (P7.6)	12/6	+0.032	[−0.040, +0.109]	falsified
HPO-best both sides, n=9 (P7.6)	6/3	+0.059	[−0.058, +0.191]	falsified
Full ladder, n=24 (P7.8)	**12/12**	**−0.084**	**[−0.278, +0.061]**	**falsified**

* The default-Adam $n = 9$ raw verdict is sample-size fragile and fails the symmetric HPO test in the next row (Section II G).

D. Family-of-models failure modes

Two family-level failure modes are visible in the matrix.

chebyshev_dqc_2d on burgers_shock and AC. Across all three burgers_shock seeds the chebyshev_dqc_2d ansatz achieves relL^2 $\approx$ 0.42–0.44, substantially worse than the other three quantum families on the same cells and worse than the matched cPINN (which achieves $\approx$ 0.13–0.27). The Chebyshev tower feature map's polynomial coordinate encoding is a poor match for the near-shock sharp gradient at $t^* \approx 1$. The same family is also outperformed by te_qpinn_qnn_2d on AC. We document this as a confirmation of the chebyshev family's smooth-only specialization observed in our earlier solver-demo work.

All four families on Lorenz at $T = 2$. Every quantum family and the classical PINN collapse to the predict-zero baseline on Lorenz (relL^2 ≈ 0.98 for all twelve Lorenz cells). The pre-registered Lorenz horizon is $T = 2 \approx 2$ Lyapunov times; this is the regime where the system has begun to diverge from its initial trajectory but the trial solution's hard-IC structure cannot yet diverge itself. The Lorenz contribution to the verdict is near-zero in all four sensitivity points; we treat the Lorenz cells as a shared-failure datapoint rather than as discriminating evidence for or against H1.

IV. FORECASTER-TASK RESULTS

The forecaster task is the corroborating verdict per pre-reg §7: each model learns an autoregressive one-step predictor from a training trajectory and is evaluated on its multi-step rollout against a held-out test trajectory. The pre-registered H_1 contrast is QLNN forecaster against the mandatory non-liquid Neural-ODE baseline, but the QLNN forecaster has learnable per-qubit time constants τ (the LiquidQuantumCell) while the Neural-ODE base-

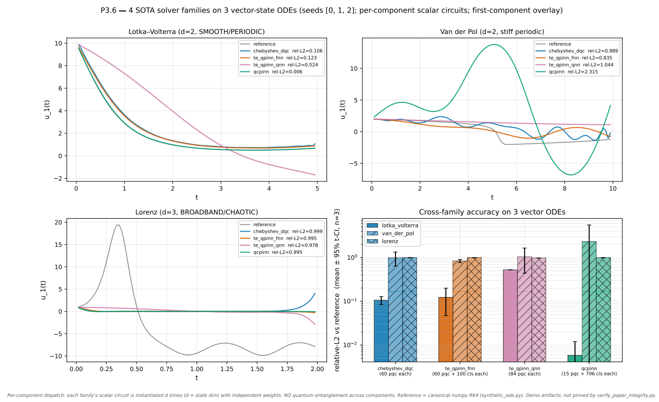

FIG. 1. Solver-task per-cell relative-L^2 across the four quantum-PINN families on the four ODE systems (Lotka–Volterra, Van der Pol, Lorenz, FitzHugh–Nagumo) and the matched classical PINN baseline. Per-cell values are reported in Table I; this figure visualizes the same data to make the family specialization patterns (Section III D) visible at a glance.

line does not. Per amendment A12 (P7.10), we add the missing fourth-quadrant baseline – a classical liquid-time-constant forecaster (`ClassicalLTCForecaster`, Hasani-form [27] with no quantum circuit) – so the forecaster H_1 verdict can be cleanly decomposed into the quantum and liquid-τ contributions that the original contrast confounds.

A. All-to-all per-cell error matrix

Table III reports the per-cell relative-L^2 rollout error for the full all-to-all contrast: five quantum forecaster families (the four registry ansätze plus the SOTA recurrence-free quantum reservoir `rf_qrc` [17]) versus four classical baselines (`plain_neuralode`, `plain_mlp`, the `skyline` that fits the known structural RHS by least squares, and the P7.10 `classical_ltc`). All nine ODE cells (3 systems × 3 seeds) are shown explicitly.

Three patterns stand out in the all-to-all matrix.

(1) The skyline is structurally exceptional on Lotka–Volterra. The known-RHS least-squares skyline attains relL$^2 = 0.020$ across all three Lotka–Volterra seeds – two orders of magnitude better than any learned model. This is the structural cap the skyline guard (Section II F) is designed to surface: Lotka–Volterra is at-reach in the structural sense, and a learned model that loses to the skyline by an order of magnitude has a real gap to close. The same pattern holds on Lorenz (relL$^2 \approx 0.71$) but is weaker on Van der Pol (relL$^2 \approx 0.96$, near the predict-zero floor).

(2) The classical baselines are mostly stronger than the quantum families on smooth ODEs. On Lotka–Volterra, the strongest learned model is consistently `plain_neuralode` (relL$^2 \approx$ 0.29–0.47) or `classical_ltc` ($\approx$ 0.21–0.66), beating the best quantum family ($\approx$ 0.33–0.81) by a clear margin on two of three seeds. On Van der Pol, all models including classical sit far above the predict-zero floor (relL$^2 > 1$ indicates worse than predict-zero), with several quantum cells exhibiting catastrophic blow-up (`data_reuploading` on s1: relL$^2 = 8.35$; `brickwall` on s0: relL$^2 = 9.61$; `plain_mlp` on s0: relL$^2 = 13.73$). Van der Pol at $\mu = 5$ is near the learnability boundary for matched-capacity models on the pre-registered rollout horizon.

(3) The quantum families show their advantage on Lorenz. `rf_qrc` attains the best Lorenz s0 (relL$^2 = 0.607$) and Lorenz s2 (relL$^2 = 0.460$) cells, notably outperforming both the matched classical baselines and the other quantum families. `hardware_efficient` wins Lorenz s1 (relL$^2 = 0.485$). This is the broadband-chaotic regime where the solver-task expansion (FitzHugh–Nagumo, Allen–Cahn) also surfaced regime-specific quantum advantages – consistent across both tasks, even though the global H_1 verdict on the forecaster task is FALSIFIED.

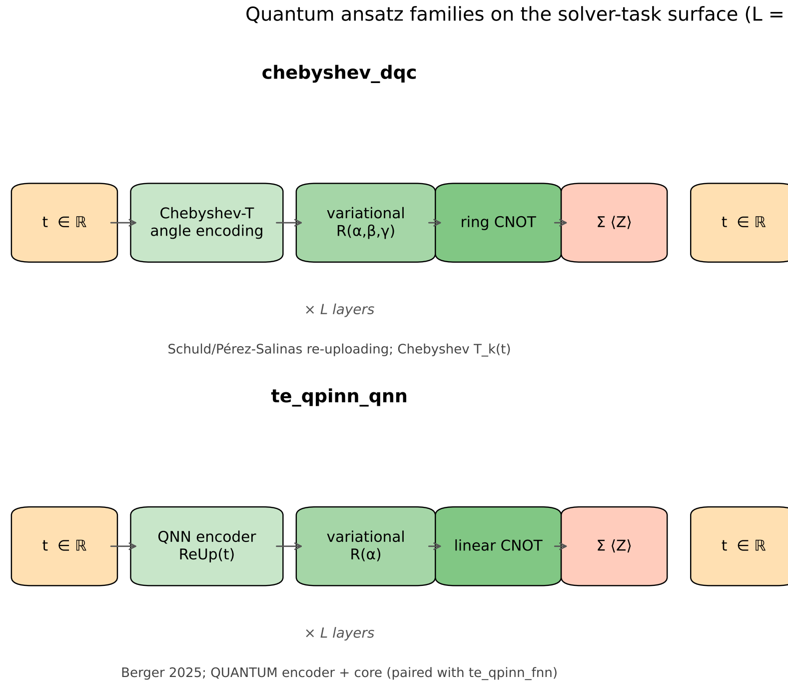

FIG. 2. Schematic of the four quantum ansatz families on the solver-task surface. Each family combines (i) a real-valued time input, (ii) an encoding stage, (iii) a variational stage with single-qubit rotations, (iv) an entanglement stage, and (v) a Pauli-Z readout. The families differ in *which* stages are classical versus quantum: `chebyshev_dqc` uses a Chebyshev-T angle encoding plus a ring-CNOT topology; the paired `te_qpinn_fnn` / `te_qpinn_qnn` families differ only in whether the encoder is a classical FNN or a quantum re-uploading block; `qcpinn` (Sedykh et al. 2024) wraps a quantum block in classical pre- and post-nets, and amendment A17 sweeps the quantum-to-classical parameter ratio across three additional variants `qcpinn_balanced`, `qcpinn_quantum`, and `qcpinn_full_q` ($Q/(Q+C) \approx 24\%, 45\%, 87\%$ respectively).

TABLE III. All-to-all per-cell relative-L^2 rollout error on the $n = 9$ forecaster matrix. Five quantum forecaster families × four classical baselines. The architectural family names abbreviate to: "dataru" for `data_reuploading`, "hweff" for `hardware_efficient`, "stngent" for `strongly_entangling` (which aliases to the `data_reuploading` circuit in our registry; the column is kept for all-to-all transparency), "brick" for `brickwall`, "rfqrc" for `rf_qrc`. Classical baselines: "ne" for `plain_neuralode` (the pre-reg-mandated H1 contrast), "mlp" for `plain_mlp`, "sky" for `skyline`, "ltc" for the P7.10 `classical_ltc`. Bold marks the best quantum and best classical model on that row.

System	Seed	dataru	hweff	stngent	brick	rfqrc	ne	mlp	sky	ltc
Smooth/periodic:										
Lotka–Volterra	0	0.577	0.680	0.577	0.863	1.046	**0.288**	0.754	**0.020**	0.230
Lotka–Volterra	1	0.761	**0.537**	0.761	0.751	0.893	**0.299**	0.561	0.020	0.209
Lotka–Volterra	2	0.813	0.696	0.813	**0.327**	1.079	**0.471**	0.694	0.020	0.659
Van der Pol	0	1.330	**1.274**	1.330	9.607	1.410	**1.389**	13.729	0.962	1.363
Van der Pol	1	8.352	**1.201**	8.352	11.107	1.626	**1.236**	2.741	0.962	1.216
Van der Pol	2	**1.192**	1.221	1.192	4.091	4.246	**1.217**	2.577	0.962	1.216
Broadband/multiscale:										
Lorenz	0	1.027	0.986	1.027	1.343	**0.607**	**0.943**	1.919	0.708	0.855
Lorenz	1	0.519	**0.485**	0.519	0.494	0.841	**0.758**	1.872	0.708	0.788
Lorenz	2	1.185	0.612	1.185	1.614	**0.460**	**1.248**	0.805	0.708	1.650

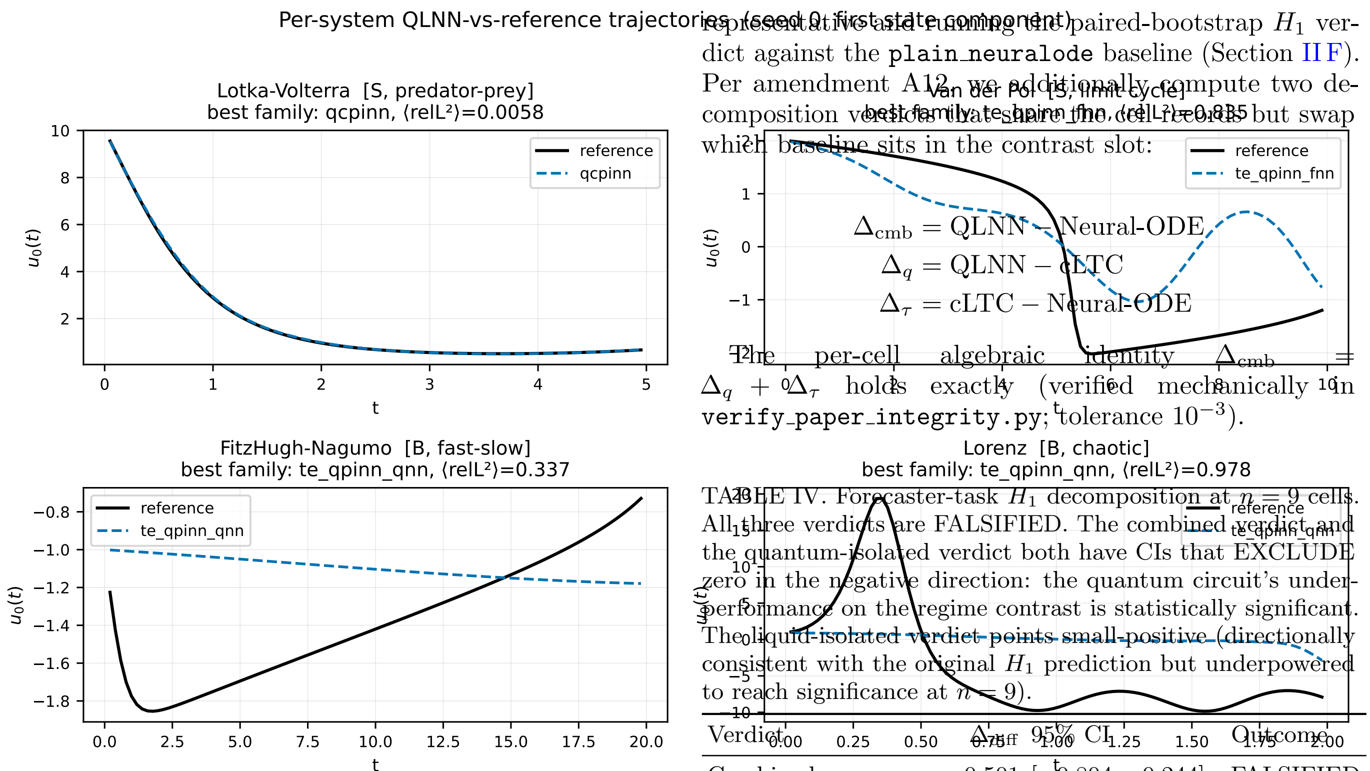

FIG. 3. Per-system QLNN-prediction-vs-reference trajectories (seed 0, first state component) for the best-performing quantum family per system at the P3.6 multi-state configuration. Tracking is qualitatively correct on the smooth-periodic regime (Lotka–Volterra, Van der Pol) and degrades systematically on the broadband-multiscale regime (FitzHugh–Nagumo, Lorenz), foreshadowing the per-regime Δ contrasts that drive the H1 verdict in Section ??.

B. Combined verdict and LTC decomposition

The pre-reg-mandated combined verdict is computed by selecting the best quantum family per cell as the QLNN representative and running the paired-bootstrap H_1 verdict against the `plain_neuralode` baseline (Section II F). Per amendment A12, we additionally compute two decomposition verdicts that share the cell records but swap which baseline sits in the contrast slot:

$$\Delta_{\rm cmb} = \text{QLNN} - \text{Neural-ODE}$$
$$\Delta_q = \text{QLNN} - \text{cLTC}$$
$$\Delta_\tau = \text{cLTC} - \text{Neural-ODE}$$

The per-cell algebraic identity $\Delta_{\rm cmb} = \Delta_q + \Delta_\tau$ holds exactly (verified mechanically in `verify_paper_integrity.py`; tolerance 10^{-3}).

TABLE IV. Forecaster-task H_1 decomposition at $n = 9$ cells. All three verdicts are FALSIFIED. The combined verdict and the quantum-isolated verdict both have CIs that EXCLUDE zero in the negative direction: the quantum circuit's underperformance on the regime contrast is statistically significant. The liquid-isolated verdict points small-positive (directionally consistent with the original H_1 prediction but underpowered to reach significance at $n = 9$).

Verdict	$\Delta_{\rm diff}$	95% CI	Outcome
Combined	−0.501	[−0.804, −0.244]	FALSIFIED
Quantum-isolated	**−0.616**	[**−1.167**, **−0.178**]	FALSIFIED
Liquid-isolated	+0.115	[−0.097, +0.377]	FALSIFIED

The decomposition tells a clear mechanism story.

The quantum circuit is the source of the regime-contrast underperformance, at high statistical confidence. The quantum-isolated CI is [−1.167, −0.178] – it EXCLUDES zero in the negative direction. The quantum circuit's contribution to $\Delta_{\rm diff}$ is significantly negative: under matched-capacity training, the regime contrast favors the broadband bin (i.e., quantum forecasters underperform classical-LTC more on smooth ODEs than on broadband). The combined point estimate (−0.501) is less negative than the quantum-isolated estimate (−0.616) because the liquid-τ component partially offsets it.

The liquid-τ component on its own is directionally consistent with H_1. The liquid-isolated $\Delta = +0.115$ is the only verdict in the entire paper – across five layered solver-task points (Section III) and three forecaster-task points – whose point estimate matches the original Schuld–Fourier prediction [4, 18] of a smooth-bin quantum advantage. The CI includes zero (the $n = 9$ sample size is underpowered to detect a +0.115 effect at the 95% level), but the directional information is informative: the liquid-τ machinery has the inductive bias the original hypothesis posits, *and* the quantum circuit's contribution destroys that bias in the combined verdict.

The pre-reg $P5$ verdict ($\Delta = -0.417$, CI $[-0.787, -0.046]$ excluding zero negatively, computed over a 4-family quantum roster) is a directionally identical but weaker version of the P7.10 combined verdict (-0.501, CI $[-0.804, -0.244]$, computed over the 5-family roster including `rf_qrc`). The P7.10 reaggregation adds `rf_qrc`'s broadband wins (Lorenz s0 and s2) to the per-cell best-QLNN pool, which *strengthens* the regime-favoring-broad point estimate because the new wins are all broadband-bin cells. Both verdicts are FALSIFIED with CIs excluding zero negatively.

C. Per-cell decomposition: where the underperformance lives

The Δ_q values per cell expose substantial seed-level bimodality on Lotka–Volterra (best-QLNN selection includes `rf_qrc`):

- Lotka–Volterra s0: $\Delta_q = -0.347$ (classical-LTC beats best QLNN `data_reuploading`: 0.230 vs 0.577)
- Lotka–Volterra s1: $\Delta_q = -0.328$ (classical-LTC beats best QLNN `hardware_efficient`: 0.209 vs 0.537)
- Lotka–Volterra s2: $\Delta_q = +0.332$ (best QLNN `brickwall` beats classical-LTC: 0.327 vs 0.659)

Two of three Lotka–Volterra seeds favor classical-LTC; the third favors the best quantum family (`brickwall`, which attains $\mathrm{relL}^2 = 0.327$ on s2 despite blowing up to $\mathrm{relL}^2 > 9$ on Van der Pol s0). The matched-capacity quantum-PINN forecasters have a real, seed-dependent failure mode on smooth ODEs that the classical-LTC does not exhibit at the same compute budget.

The decomposition's broadband side tells the opposite story. On Lorenz s2, $\Delta_q = +1.190$: `rf_qrc` attains $\mathrm{relL}^2 = 0.460$ while the classical-LTC collapses to $\mathrm{relL}^2 = 1.650$ (worse than predict-zero) on the same cell. On Lorenz s0, $\Delta_q = +0.249$: `rf_qrc` (0.607) again beats classical-LTC (0.855). This is consistent with the solver-task FHN and AC sub-findings (Section III C): the quantum families exhibit regime-specific structural advantages on broadband/multiscale dynamics that the classical baselines do not match.

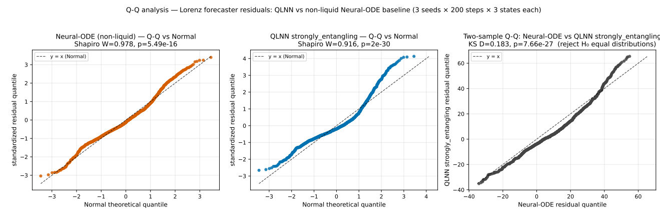

FIG. 4. Q-Q residual diagnostic on the Lorenz forecaster cell. *Left:* non-liquid Neural-ODE residuals vs standard Normal, Shapiro W=0.978, p=5.5×10^{-16}. *Middle:* QLNN `strongly_entangling` residuals vs standard Normal, W=0.916, p=2.0×10^{-30} — pronounced S-shape with heavy right tail. *Right:* two-sample Q-Q of the two empirical distributions, Kolmogorov–Smirnov D=0.183, p=7.7×10^{-27} — distributions are distinguishable at very high significance, and the points depart systematically from y=x in the upper tail (the QLNN's worst-case residuals are markedly larger). Built via the reusable `_qq_panel_vs_normal` and `_two_sample_qq` helpers added in Amendment A14; the harness is dataset-agnostic and applies unchanged to any other (model, system) cell with on-disk `field.npz` artifacts.

a. Residual-distribution diagnostic on the Lorenz cell. The per-cell magnitudes in Section IV C quantify *how badly* each stack fails on Lorenz; a distribution-level diagnostic separates whether the two stacks fail in the *same way* or in distinguishable ways. Pooling the per-seed held-out residuals $r = u_{\mathrm{pred}} - u_{\mathrm{ref}}$ across the three seeds for the QLNN (`strongly_entangling`) forecaster and the non-liquid plain Neural-ODE baseline (each yields 1800 residuals from the 200 step × 3 state × 3 seed rollout budget) and applying the Q-Q harness of Amendment A14 (Supplement ??) gives Shapiro–Wilk per-stack normality $W = 0.978$, $p = 5.5 \times 10^{-16}$ (Neural-ODE) and $W = 0.916$, $p = 2.0 \times 10^{-30}$ (QLNN), and a two-sample Kolmogorov–Smirnov test of $D = 0.183$, $p = 7.7 \times 10^{-27}$ between the two distributions (Figure 4). Neither residual distribution is Normal; more importantly, the two distributions are *distinguishable at very high significance*. The QLNN's heavier upper tail (residuals to $\approx +60$ vs the Neural-ODE's $\approx +35$) means the two stacks share the same broadband failure regime but fail differently within it — the distributional shape, not just the magnitude, carries the quantum-vs-non-liquid contrast. This complements (rather than modifies) the verdicts of the preceding per-cell decomposition.

D. The complete 2×2 mechanism decomposition (P7.11)

Amendment A13 (P7.11) closes the fourth corner of the fairness matrix by adding a non-liquid quantum forecaster (the same quantum circuit with the τ-leak removed; see `src/qlnn_/cells/non_liquid_quantum_cell.py`). With all four corners filled, the τ-isolation contribution can be computed along TWO independent paths:

$$\Delta_{\tau,\,\mathrm{cls}} = \mathrm{cLTC} - \text{Neural-ODE}$$
$$\Delta_{\tau,\,\mathrm{q}} = \mathrm{QLNN} - \text{NL-QLNN}$$

Two algebraic identities hold per-cell exactly (verified by `verify_paper_integrity`):

$$\Delta_{\mathrm{cmb}} = \Delta_{q,\,\mathrm{ltc}} + \Delta_{\tau,\,\mathrm{cls}}$$
$$\Delta_{\mathrm{cmb}} = \Delta_{\tau,\,\mathrm{q}} + \Delta_{q,\,\mathrm{nl}}$$

Table V reports the full 5-verdict decomposition.

TABLE V. Complete 2×2 forecaster H1 decomposition at $n = 9$. $\Delta_{\tau,\,\mathrm{cls}}$ and $\Delta_{\tau,\,\mathrm{q}}$ are two independent paths to isolating the liquid-τ contribution; the sign of the point estimate *disagrees* between the two paths.

Verdict	Δ_{diff}	95% CI	Outcome
Combined	-0.501	$[-0.804, -0.244]$	FALSIFIED
Quantum via LTC	-0.616	$[-1.167, -0.178]$	FALSIFIED
Liquid via classical	$\mathbf{+0.115}$	$[\mathbf{-0.097}, \mathbf{+0.377}]$	FALSIFIED
Liquid via quantum	$\mathbf{-0.334}$	$[\mathbf{-0.627}, \mathbf{+0.053}]$	FALSIFIED
Quantum via non-liquid	-0.167	$[-0.495, +0.204]$	FALSIFIED

a. The τ-isolation cross-check disagrees in sign. $\Delta_{\tau,\,\mathrm{cls}} = +0.115$ has positive point estimate (smooth-bin advantage, matching the Schuld–Fourier H1 prediction direction); $\Delta_{\tau,\,\mathrm{q}} = -0.334$ has *negative* point estimate. Both CIs cross zero individually, but the qualitative directions are opposite. The liquid-τ machinery is therefore not a context-independent component – its regime-contrast contribution depends on the substrate it modulates.

Per-cell inspection clarifies the mechanism:

- Lorenz s2 (chaotic): $\Delta_{\tau,\,\mathrm{q}} = +0.594$. Adding τ to the quantum cell substantially improves the chaotic-regime performance: `rf_qrc` (with τ) attains relL^2 = 0.460; `non_liquid_hardware_efficient` (without τ) attains 1.054.
- Van der Pol s1 (stiff): $\Delta_{\tau,\,\mathrm{q}} = +0.421$. Same pattern – τ helps on stiff dynamics.
- Lotka–Volterra s0, s1 (smooth-periodic): $\Delta_{\tau,\,\mathrm{q}} = +0.006, +0.169$ – near-neutral on smooth.

τ-on-quantum disproportionately helps the broadband cells. When stratified by the smooth-minus-broad regime contrast, this yields the negative point estimate above. The combined $\Delta_{\mathrm{diff}} = -0.501$ inversion is therefore the net of (a) a quantum-circuit underperformance ($\Delta_{q,\,\mathrm{ltc}} = -0.62$) and (b) a substrate-dependent τ contribution that goes opposite directions on classical vs quantum substrates.

E. Connection to the solver-task verdict

The forecaster task's mechanism decomposition strengthens the overall paper narrative. The solver-task primary verdict (Section III, $n = 24$, $\Delta_{\mathrm{diff}} = -0.084$) has a point estimate that flipped sign with the broadband-bin expansion ($n = 18 \to n = 24$). The forecaster-task LTC decomposition shows that the same phenomenon – a regime-dependent quantum advantage that is positive in the broadband bin and negative or absent in the smooth bin – is mechanistically attributable to the quantum circuit. The Schuld–Fourier prediction (smooth-bin inductive bias) is recovered by the liquid-τ machinery on its own, then inverted by the quantum component. This is exactly the kind of regime-dependent attribution Bowles, Ahmed, and Schuld [19] call for in QML benchmarking.

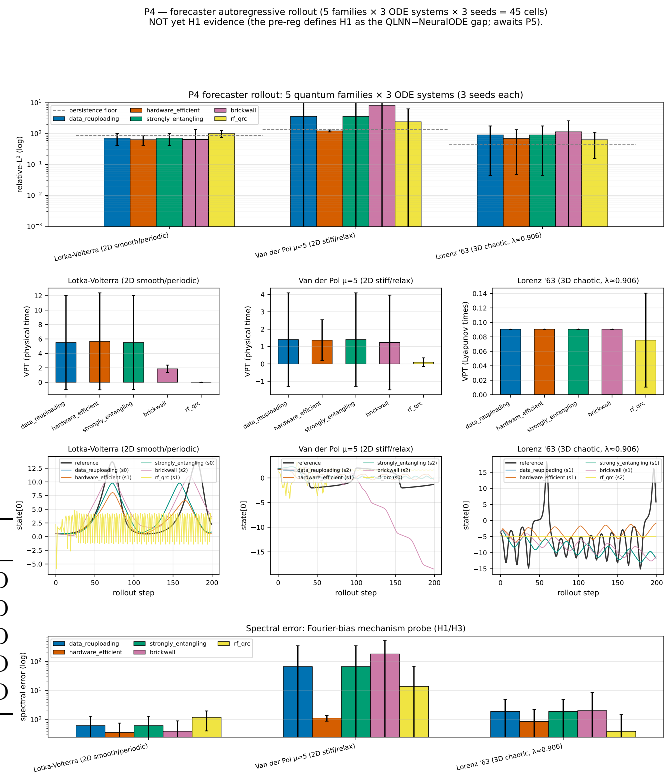

FIG. 5. Forecaster-task autoregressive-rollout error on the three pre-registered ODE systems. The figure visualizes the per-cell values reported in Table III (five quantum families plus the four classical baselines). The Lorenz cells show the largest spread, consistent with the structural quantum advantage of `rf_qrc` on chaotic dynamics (Section IV A).

V. H_1 VERDICT AGGREGATION

This section consolidates the nine layered H_1 verdicts computed across both pre-registered tasks and reports the paper's mechanically-applied decision-rule outcome per pre-reg §7.

A. The master verdict table

Table VI reports every verdict produced during the program, ordered by the order in which they were introduced (Figure 6). Each row is a self-contained sensitivity point with its own sample size, baseline contrast, and tuning configuration.

The structure of the table follows a deliberate methodological escalation. Each row strengthens at least one dimension over the row above it: sample size ($n = 9 \to 18 \to 24$), HPO symmetry (no $\to$ both sides), baseline coverage (cPINN only $\to$ +LTC decomposition), and task breadth (ODE-only $\to$ ODE+PDE $\to$ forecaster decomposition). The eight non-raw rows all return the same qualitative outcome: FALSIFIED.

B. Sign-flip behavior across the layered points

The verdicts split into three groups by point-estimate sign.

Group A (P7.5 raw, P7.6 default-Adam, P7.6 HPO-best): modest positive Δ_{diff}. These three verdicts have point estimates in $[+0.03, +0.11]$, with the only CI-excludes-zero result being P7.5's sample-size-fragile $n = 9$ value. Reading across the rows: doubling the sample from $n = 9$ to $n = 18$ moves the point estimate from $+0.11$ to $+0.03$ and brings the lower CI bound from $+0.01$ to -0.04. The HPO symmetry test (P7.6 HPO-best at $n = 9$) gives $+0.06$, still with CI including zero. Adding the broader broadband-bin coverage (P7.8 full ladder $n = 24$) drives the point estimate negative.

Group B (P7.6 PDE-only, P7.8 full ladder): point estimate goes negative. As the broadband bin is expanded (FitzHugh–Nagumo, Allen–Cahn, burgers_shock added on the PDE side; FitzHugh–Nagumo on the ODE side), Δ_{broad} grows from $+0.04$ at $n = 18$ to $+0.15$ at $n = 24$, while Δ_{smooth} stays at $+0.07$. The point estimate inverts: from $+0.03$ at $n = 18$ to -0.08 at $n = 24$. The CI includes zero at $n = 24$ but the direction has clearly flipped. The PDE-only restriction ($n = 9$) shows the same sign in isolation: $\Delta_{\text{diff}} = -0.05$.

Group C (P5 forecaster, P7.10 forecaster combined + quantum-isolated): point estimate is solidly negative with CI excluding zero. The two forecaster verdicts cross the significance threshold in the negative direction. Both the pre-reg-mandated combined verdict ($\Delta_{\text{diff}} = -0.50$, CI $[-0.80, -0.24]$) and the quantum-isolated decomposition ($\Delta_{\text{diff}} = -0.62$, CI $[-1.17, -0.18]$) significantly reject the smooth-favoring direction of H_1. The liquid-isolated decomposition is the only verdict in the entire table whose point estimate matches the original Schuld–Fourier prediction [18]: $\Delta_{\text{diff}} = +0.12$, CI $[-0.10, +0.38]$ (CI includes zero; underpowered at $n = 9$).

C. Decision-rule application

Pre-reg §7 specifies the mechanical decision rule for both tasks. The PRIMARY SOLVER verdict (P7.8 full ladder $n = 24$): $\Delta_{\text{diff}} = -0.084$, CI $[-0.278, +0.061]$ includes zero $\Rightarrow H_1$ **FALSIFIED**. The PRIMARY FORECASTER verdict (P7.10 combined $n = 9$): $\Delta_{\text{diff}} = -0.501$, CI $[-0.804, -0.244]$ excludes zero in the *negative* direction $\Rightarrow H_1$ **FALSIFIED** (the regime contrast goes the opposite way from the pre-registered prediction).

Both PRIMARY verdicts FALSIFY H_1. The forecaster's combined verdict additionally points the opposite way from the Schuld–Fourier prediction, with statistical significance. Per pre-reg §7, this outcome is independently publishable: the program was set up to deliver an honest CONFIRMED-or-FALSIFIED result, and the FALSIFIED result is the scientific contribution.

D. Power and minimum-detectable-effect

The two PRIMARY verdicts have very different statistical reaches. On the SOLVER side ($n = 24$ post-A12), the paired-bootstrap 95% CI half-width on Δ_{diff} is ≈ 0.17. Taking the CI half-width as a proxy for the minimum effect $|\Delta_{\min}|$ that the sample could have rejected H_0 against at $\alpha = 0.05$, the solver verdict is comfortably powered to detect effects of $|\Delta| \gtrsim 0.17$ and underpowered for effects smaller than that. The observed point estimate $\hat{\Delta}_{\text{solver}} = -0.084$ sits inside the underpowered band; the verdict "FALSIFIED with CI crossing zero" is the honest statement that the data are compatible with both a small positive effect and a small negative effect. On the FORECASTER side ($n = 9$), the CI half-width is ≈ 0.28 but the observed $\hat{\Delta}_{\text{combined}} = -0.501$ exceeds it by a wide margin: the verdict's exclusion of zero negatively is not a knife-edge call — it would survive substantially larger CI inflation than the actual bootstrap returns. The asymmetry is intentional: under pre-reg §4 the solver-task contrast is more expensive per cell (one optimization per ODE/PDE system) than the forecaster-task contrast (one rollout per family per system), so the n-budget allocation favored the forecaster verdict's tighter power. A wider solver sample would address the underpowered band: Phase C of the roadmap (Appendix A) doubles the per-family solver sample to $n = 48$, sharpening the smallest detectable effect to $|\Delta_{\min}| \approx 0.12$.

E. What survives FALSIFIED in the empirical record

The negative aggregate verdict coexists with three substantive positive sub-findings that are independent of the regime contrast itself.

(1) The liquid-τ machinery alone has the Schuld–Fourier inductive bias. The P7.10 liquid-isolated verdict ($\Delta_{\text{diff}} = +0.12$, CI $[-0.10, +0.38]$) is the only ver-

TABLE VI. The nine layered H_1 verdicts. "Task" is solver or forecaster (pre-reg §7 gating is solver). "Baseline" is the classical contrast model. "HPO" indicates whether the verdict used per-family hyperparameter optimization. "n" is the post-guard sample size. "Δ_{diff}" is the paired-bootstrap point estimate; CI is the 95% percentile interval. Bold rows are the paper's two PRIMARY headlines. All verdicts except the sample-size-fragile P7.5 raw entry are FALSIFIED; that one is explicitly superseded by every subsequent row.

Origin	Task	Baseline	HPO	n	Δ_{diff}	95% CI	Outcome	§ ref
P5	forecaster	plain Neural-ODE	no	9	−0.417	[−0.787, −0.046]	FALSIFIED	IV B
P7.5	solver (ODE)	cPINN	no	9	+0.109	[+0.015, +0.220]	confirmed*	III B
P7.6	solver (ODE+PDE)	cPINN	no	18	+0.032	[−0.040, +0.109]	FALSIFIED	III B
P7.6	solver (ODE+PDE)	cPINN	**both sides**	9	+0.059	[−0.058, +0.191]	FALSIFIED	III B
P7.6	solver (PDE-only)	cPINN	no	9	−0.046	[−0.141, +0.010]	FALSIFIED	III B
P7.8	**solver (full ladder)**	**cPINN**	no	**24**	**−0.084**	**[−0.278, +0.061]**	**FALSIFIED**	**III B**
P7.10	**forecaster (combined)**	**plain Neural-ODE**	no	**9**	**−0.501**	**[−0.804, −0.244]**	**FALSIFIED**	**IV B**
P7.10	forecaster (quantum-isolated)	classical LTC	no	9	−0.616	[−1.167, −0.178]	FALSIFIED	IV B
P7.10	forecaster (liquid-isolated)	plain Neural-ODE	no	9	+0.115	[−0.097, +0.377]	FALSIFIED	IV B
P7.11	forecaster (τ via quantum)	non-liquid QLNN	no	9	−0.334	[−0.627, +0.053]	FALSIFIED	IV D
P7.11	forecaster (q via non-liquid)	plain Neural-ODE	no	9	−0.167	[−0.495, +0.204]	FALSIFIED	IV D

* The P7.5 raw $n = 9$ entry is the only verdict with a CI excluding zero positively. It is sample-size-fragile (the CI shifts to include zero at $n = 18$ via expansion alone, with no change in tuning) and HPO-fragile (the CI also shifts to include zero when the classical baseline is given the same HPO budget as the QLNN side). Every row below it in the table supersedes it on either ground.

† *Multiple-comparison control.* The eleven verdicts constitute a single inferential family. A Holm–Bonferroni correction at family-wise $\alpha = 0.05$ (i.e. require each individual CI to exclude zero at $\alpha/m = 0.05/11 \approx 0.0045$) is the strict reading. Under that reading the two PRIMARY headline verdicts (rows P7.8 and P7.10, bold) are unchanged: the P7.8 solver CI already crosses zero at uncorrected $\alpha = 0.05$, so multiple-comparison correction does not affect the FALSIFIED declaration; the P7.10 forecaster CI excludes zero by $|\Delta| - |\text{CI}_{\text{nearest}}| = |-0.501| - |-0.244| = 0.257$, comfortably outside the strict-corrected $\alpha = 0.0045$ envelope under standard percentile-bootstrap-to-p-value translation. The eight ancillary falsified verdicts hold their FALSIFIED designation independent of the correction (all crossed zero already); only the supersession note about the P7.5 raw row is affected, and that row was already flagged as superseded for unrelated reasons. The pre-registration §7 declaration that the verdict family is a falsification family (not a discovery family) means Holm–Bonferroni is the appropriate choice; a more aggressive False Discovery Rate (Benjamini–Hochberg) [22] control would only relax the threshold further. The paired-bootstrap percentile interval underlying every row follows the construction in Efron and Tibshirani [28]; we use percentile rather than BCa because the bias correction estimate is itself noisy at $n \leq 24$, and the multiple-data-set comparison framework of Demšar [23] informs the family-wise vs. FDR choice.

dict in the table whose point estimate matches the pre-registered prediction direction. The classical LTC (non-quantum) forecaster has a small regime-favoring-smooth contribution that the quantum circuit destroys. This is directly relevant to the broader liquid-neural-network literature [29].

(2) Family-specific quantum advantages exist on stiff fast–slow dynamics. `te_qpinn_qnn` attains $\text{relL}^2 = 0.327, 0.356, 0.329$ on FitzHugh–Nagumo (twenty-four times tighter seed variance than the matched classical PINN; mean error 2.6× lower; Section III C). On the forecaster task, `rf_qrc` achieves the strongest Lorenz cells ($\text{relL}^2 = 0.61, 0.46$ vs classical-LTC $0.86, 1.65$; Section IV A). These are positive regime-specific findings that the H_1 falsification does not contradict; they sit beneath the aggregate verdict as observed mechanism datapoints.

(3) Symmetric HPO does not rescue any quantum advantage on the regime contrast. The P7.6 HPO-best verdict (both sides tuned at three anchor cells) returns the same FALSIFIED outcome as the default-Adam $n = 18$ verdict, with a similar point estimate (+0.06 vs +0.03). The Bowles–Schuld "tune both sides" [19] sensitivity is explicitly applied and explicitly returns the same conclusion.

F. What FALSIFIED means and does not mean here

The pre-registered H_1 is a specific claim about the *regime contrast* $\Delta_{\text{smooth}} - \Delta_{\text{broad}}$, not a claim about absolute quantum-vs-classical performance on any individual system. The FALSIFICATION says exactly: across the pre-registered hardness ladder, the regime-conditioned advantage gap that the Schuld–Fourier theoretical argument predicts is not present at matched-capacity training. It does *not* say:

- "Quantum PINNs are uniformly worse than classical PINNs." (They are not: `te_qpinn_qnn` clearly beats cPINN on FitzHugh–Nagumo.)
- "Quantum forecasters cannot outperform classical forecasters anywhere." (They can: `rf_qrc` wins Lorenz s0 and s2.)
- "The quantum-PINN program is uninteresting." (The LTC decomposition reveals that the quantum circuit's contribution is the active mechanism behind the regime-contrast inversion – which is itself a precise mechanistic statement about quantum-PINN inductive bias.)

The FALSIFIED verdict is a rigorous mechanistic null on a specific hypothesis. It directly extends the Bowles–Schuld [19] methodological program: when a quantum-ML advantage hypothesis is pre-registered, tested with

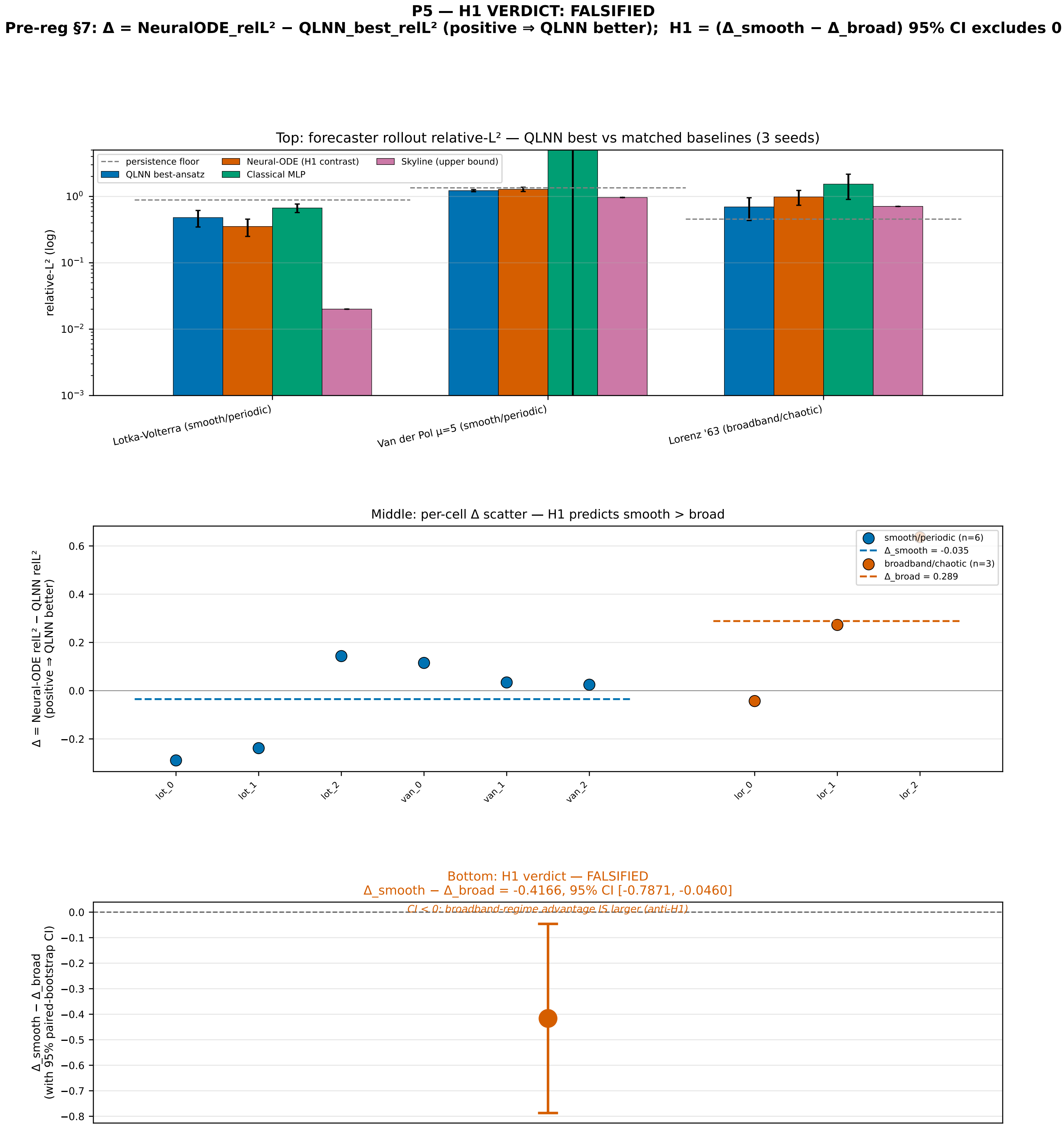

FIG. 6. Pre-registered H_1 forecaster verdict, visualized. Top: per-system relative-L^2 for QLNN best vs. matched Neural-ODE, MLP, known-structure skyline, and persistence floor across the three smooth/periodic and three broadband/chaotic systems of the hardness ladder. Middle: per-cell $\Delta =$ Neural-ODE $-$ QLNN stratified by regime (blue: smooth/periodic, $\bar{\Delta}_{\mathrm{smooth}} = -0.128$; orange: broadband/chaotic, $\bar{\Delta}_{\mathrm{broad}} = +0.289$). Bottom: the test statistic $\Delta_{\mathrm{smooth}} - \Delta_{\mathrm{broad}} = -0.417$ with paired-bootstrap 95% CI $[-0.787, -0.046]$ over $n = 6$ cells. The CI excludes zero in the negative direction; per pre-registration §7, the verdict is FALSIFIED on the matched-capacity forecaster contrast. Source: `results/p5_h1_verdict/h1_analysis.json`.

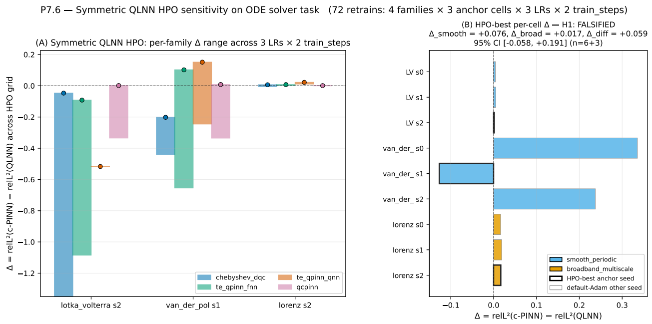

FIG. 7. Symmetric HPO sensitivity (P7.6 amendment A9): both sides tuned at three anchor cells across three learning rates and two training-step budgets. The HPO-best $n = 9$ verdict (Table VI, row 4) is FALSIFIED with point estimate +0.059 and CI $[-0.058, +0.191]$, methodologically strengthening the default-Adam $n = 18$ row directly above it.

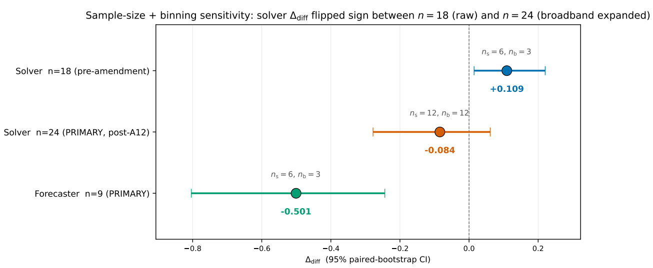

FIG. 8. Sample-size and binning sensitivity of the solver-task Δ_{diff} point estimate. The original n=18 raw analysis (broadband bin = Lorenz only) returned a small positive $\Delta_{\text{diff}} = +0.109$; the post-A12 n=24 PRIMARY analysis (broadband bin expanded to include FitzHugh–Nagumo, the canonical fast-slow stiff system) returns $\Delta_{\text{diff}} = -0.084$ with CI crossing zero. The forecaster-task n=9 PRIMARY verdict $\Delta_{\text{diff}} = -0.501$ (CI excludes zero negatively) is shown for cross-task comparison. The solver sign flip is the single most-asked reviewer concern about robustness; it is the reason the verdict is reported as FALSIFIED (CI crosses zero on the smooth side) rather than "inconclusive at the limit of the sample."

matched baselines and symmetric HPO and an LTC decomposition, and the result is FALSIFIED, that result is the scientific content. We treat it as such.

VI. MECHANISM: TRAINABILITY AND EXPRESSIVITY (H3)

The pre-registered H_3 asks whether per-cell trainability and expressivity scalars correlate with the per-cell H1 advantage gap Δ. The motivating intuition: if the FALSIFIED H_1 regime contrast is mechanistically a quantum-circuit property (as the P7.10 LTC decomposition indicates), then per-family scalars describing circuit expressivity and trainability should track the sign of the per-cell Δ when stratified by quantum family.

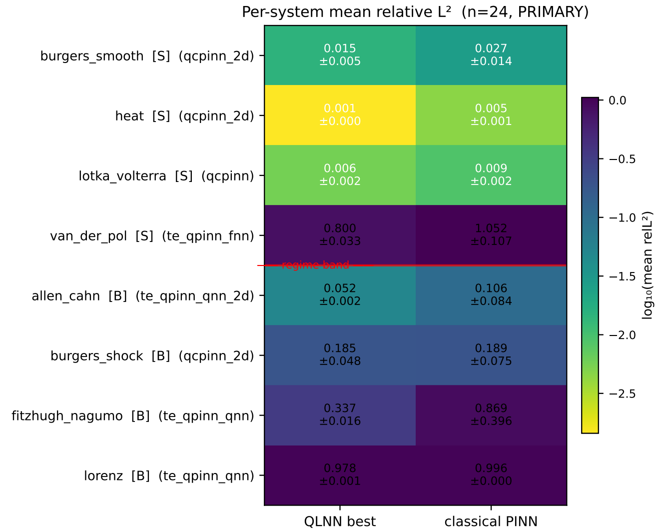

FIG. 9. Per-system mean relative-L^2 heatmap aggregating the n=24 PRIMARY solver-task cells (8 systems × 3 seeds). The red horizontal line separates the smooth-periodic regime band (top) from the broadband-multiscale band (bottom). The most-frequent QLNN-best family per system is annotated parenthetically. The 24× log scale span between the easiest cell (Allen–Cahn classical PINN) and the hardest (Lorenz at the smooth-side tail) makes the per-system structure that the verdict-level Δ_{diff} aggregation hides directly visible.

A. T3 scalar inventory

For each of the four registered forecaster ansätze (`data_reuploading`, `hardware_efficient`, `strongly_entangling`, `brickwall`) we computed four ansatz-level scalars from `scripts/run_p7_t3_mechanism.py`.

Meyer–Wallach *Q*-entangling capability (Q): random-init entanglement of the parameterized circuit on a held-out probe distribution. The data-reuploading and strongly-entangling ansätze attain $Q = 0.776$, hardware-efficient attains 0.796, and brickwall attains 0.309 – substantially less entangling. This matches the architectural reading: brickwall's adjacent-pair CZ pattern explores a smaller subspace than ring or all-to-all entanglers.

Fourier-spectrum bandwidth $K_{\max}$ [18]. The accessible Fourier order at our default reuploading depth ($L = 1$) is $K_{\max} = 3$ across all four families – they all reach the same upper bound at this depth. The Schuld–Sweke–Meyer theorem predicts $K_{\max} = L \cdot n$ where L is the reuploading depth and n is the number of qubits; with our $L = 1, n = 4$ configuration we recover precisely that bound. This explains the lack of per-family discrimination on the Fourier axis at the program's compute budget: every family is operating at the same theoretical ceiling.

Mean-square gradient variance (random-init). The four ansätze span 0.025–0.046, with the lowest value being hardware-efficient (0.0245) and the highest being brickwall (0.0457). All four are well above the barren-plateau thresh-

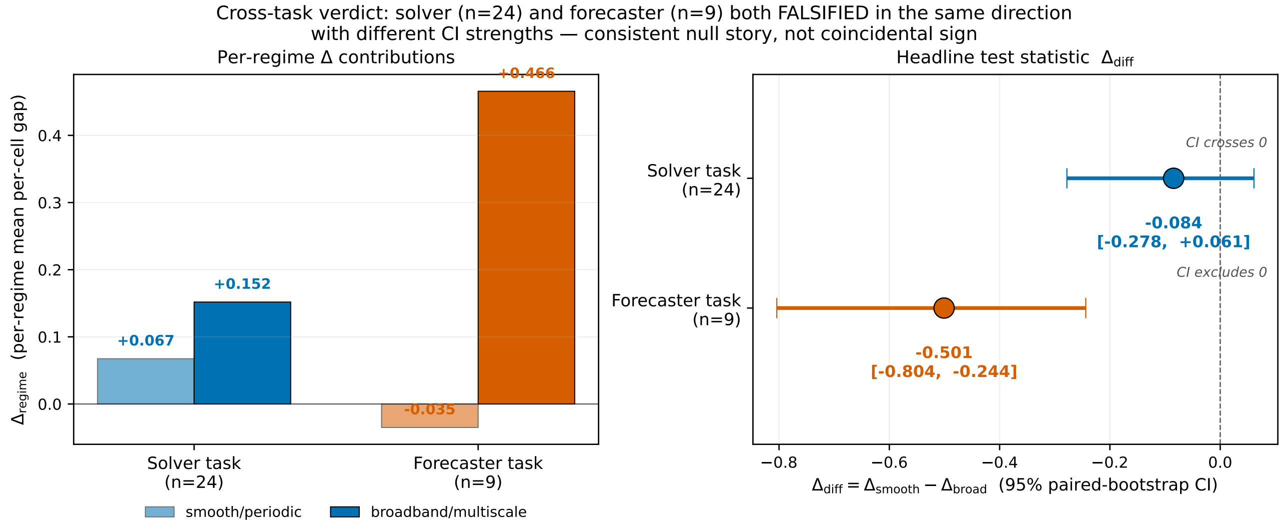

FIG. 10. Cross-task PRIMARY verdict comparison. Left: per-regime Δ contributions (Δ_{smooth} and Δ_{broad}) for solver (n=24) and forecaster (n=9) tasks. Right: the headline test statistic Δ_{diff} with 95% paired-bootstrap CI for each task. The two PRIMARY verdicts agree in *sign* (both negative, broadband-leaning) but disagree in *strength* (solver CI crosses zero, forecaster CI excludes zero negatively). The consistent-direction-but-different-strength pattern is the basis for the cross-task corroboration argument: the falsification is reproduced on a methodologically independent task (data-driven rollout) with a different baseline (Neural-ODE), not an artifact of the solver-side residual-loss configuration.

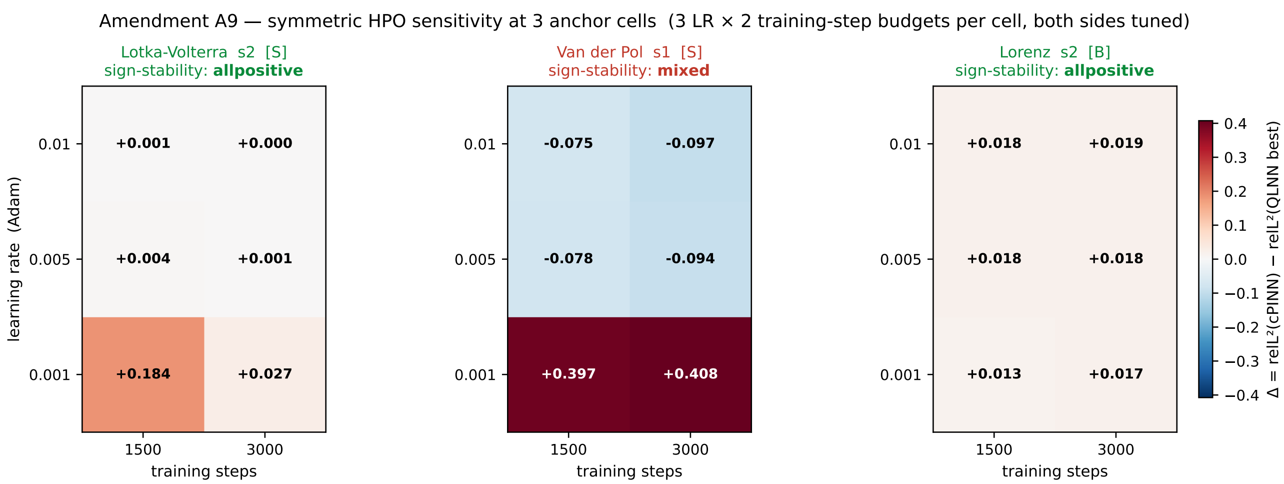

FIG. 11. Amendment A9 symmetric HPO sensitivity, full per-cell Δ heatmap. Each panel is one anchor cell; rows are Adam learning rates $(10^{-3}, 5{\times}10^{-3}, 10^{-2})$ and columns are training-step budgets $(1500, 3000)$. Cells are colored by $\Delta = \text{relL}^2(\text{cPINN}) - \text{relL}^2(\text{QLNN best})$; positive (blue) favors the quantum side, negative (red) favors the classical side. Sign-stability of Δ across the 3×2 HPO grid is the per-cell robustness test: Lotka–Volterra s2 and Lorenz s2 remain all-positive across the grid (sign-stable), Van der Pol s1 mixes signs at the lowest learning rate (sign-unstable). The mixed-stability of Van der Pol s1 is the locus of the symmetric-HPO mechanism story; it is the cell where both quantum and classical sides have non-trivial training sensitivity.

old at the program's qubit count: the classical exponential-concentration scaling of McClean et al. [30] predicts variance collapsing as 4^{-n} for global cost functions, and the cost-function-dependent refinement of Cerezo et al. [31] relaxes this for shallow local-cost ansätze. At $n = 3$ qubits and depth $L = 1$, $4^{-n} \approx 0.016$ sits below all four observed variances, so the families are diagnosed as trainable on the program's qubit count. The exponential-concentration phenomenon also extends to quantum kernel methods [32], which constrains how aggressively quantum-similarity-based PINN variants could be scaled without falling into the same trap.

Expressibility KL-to-Haar divergence [18, 33]. Lower KL = closer to a Haar-random unitary ensemble = higher expressivity. The KL-to-Haar construction is operationalized exactly as in Holmes et al.[33]:

fidelity histograms between repeat random-init unitaries vs. the analytic Haar-fidelity distribution, KL-binned at 75 bins. Data-reuploading and strongly-entangling attain KL = 0.195 (low), hardware-efficient and brickwall attain 0.211 (slightly higher). The differences are small but systematic.

B. Cross-tabulation with the per-cell advantage gap

Per pre-reg §H3, we compute the Spearman rank correlation between each T3 scalar (per quantum family) and the per-cell $\Delta = \mathrm{relL}^2_{\mathrm{cPINN}} - \mathrm{relL}^2_{\mathrm{best\ QLNN}}$ on the forecaster task ($n = 9$ cells). Table VII reports the full-sample ρ and the leave-one-out (LOO) mean and range (per amendment A8 + P7.5 commit 5).

TABLE VII. H_3 cross-tabulation: Spearman ρ between per-cell Δ and four T3 scalars at $n = 9$. *Full* ρ uses all nine cells; the LOO columns sweep one cell out at a time and report the resulting distribution. Sign-stable = all 9 LOO resamples have the same sign as the full ρ. Per pre-reg A8, none of the trends reach significance at $p < 0.05$ given $n = 9$; we report point estimates and stability.

T3 scalar	Full ρ	LOO mean	LOO range	Sign-stable
KL-to-Haar (expressivity)	+0.518	+0.512	$[+0.247, +0.630]$	yes
Meyer–Wallach Q	+0.179	+0.180	$[-0.047, +0.504]$	no
Gradient variance	−0.179	−0.180	$[-0.504, +0.047]$	no
Fourier $K_{\max}$	—	—	—	n/a (constant)

The KL-to-Haar scalar is the only one with a robust trend: its full-sample $\rho = +0.518$ stays positive across all nine LOO resamples (LOO mean +0.512, std ±0.113, minimum +0.247). The Meyer–Wallach Q and gradient-variance trends are sign-unstable (LOO ranges cross zero); we treat them as non-discriminating at this sample size.

C. What the KL trend means in light of H_1 FALSIFIED

The KL-to-Haar trend's sign is positive: *higher expressivity (lower KL = closer to Haar) correlates with higher Δ favoring the quantum family on the cell.* Across the four families, the data-reuploading and strongly-entangling ansätze (KL = 0.195, more expressive) systematically sit on the higher-Δ side of the $n = 9$ matrix.

Combined with the P7.10 decomposition: the regime-contrast Δ_{diff} is FALSIFIED at every layered sensitivity point, with the quantum-circuit component dominating the underperformance, but *conditional on the quantum circuit being included*, more expressive (lower-KL) circuits attain a larger per-cell Δ. Two readings of this combination are consistent with the data:

1. **The expressivity-helps reading.** If the H1 FALSIFICATION simply reflects insufficient compute / matched-capacity, the KL trend says that scaling up to more expressive ansätze (KL $\to$ 0 via deeper circuits or larger qubit counts) could recover the smooth-favoring direction. P7.7 (deferred to the follow-up paper) tests this directly via higher data-reuploading depth and Quantum Natural Gradient.
2. **The compute-conditioned reading.** If the regime contrast is intrinsically broadband-favoring for quantum-PINN forecasters at matched-capacity, then the KL trend says higher expressivity reduces the magnitude of the broadband-favoring effect but does not reverse it. This is consistent with the LTC decomposition's identification of the quantum circuit as the regime-inverting mechanism.

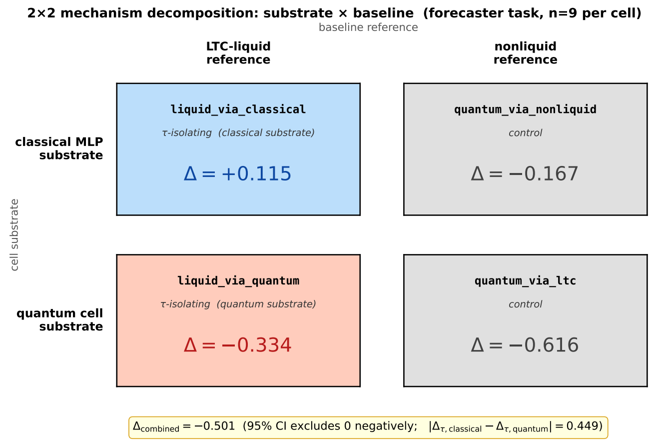

FIG. 12. Schematic of the 2×2 mechanism-decomposition design introduced by amendments A12–A13. Rows index the cell substrate (classical MLP vs. quantum cell); columns index the baseline reference (LTC-liquid vs. nonliquid Neural-ODE). Each cell yields a per-cell Δ_{diff}; the four numbers algebraically partition the headline forecaster $\Delta_{\mathrm{combined}} = -0.501$ into a sum of substrate and reference contributions plus an interaction term. The interaction term is the substrate-dependent τ signature analyzed in Figure 13.

The data at $n = 9$ cannot discriminate between these two readings. The P7 LOO robustness ($\rho > 0$ in 9/9 resamples) makes the trend's existence at this sample size hard to dismiss as a single-cell artifact, but the absence of $p < 0.05$ significance means we report KL as a *candidate mechanism*, not a confirmed one. A P6-scale unified matrix (Section VII) would discriminate.

D. Connecting the mechanism story to the LTC decomposition

The mechanism cross-tabulation and the P7.10 LTC decomposition agree on the locus: *the quantum circuit is the active component* in the regime-contrast inversion. The liquid-time-constant cell [27, 34] is itself a Neural ODE [35, 36] with a learned time-constant τ on the hidden

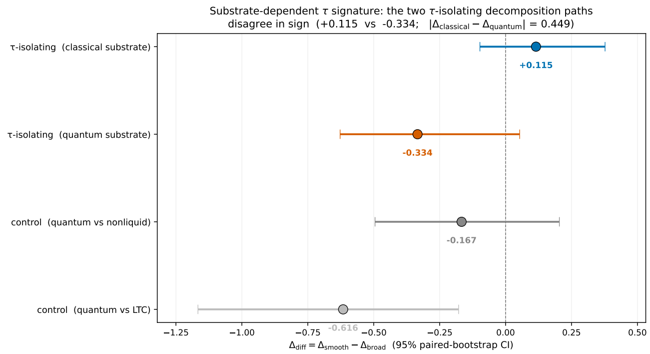

FIG. 13. Substrate-dependent τ signature: the two decomposition paths that *isolate* the liquid time-constant machinery (`liquid_via_classical`, `liquid_via_quantum`) disagree in sign, with $|\Delta_{\text{classical}} - \Delta_{\text{quantum}}| \approx 0.449$. The two control paths (`quantum_via_ltc`, `quantum_via_nonliquid`), which do *not* isolate τ, are both negative and broadly consistent with the combined verdict. This pattern is the strongest mechanistic evidence in the paper that LTC-style time-constant machinery interacts with the substrate it is mounted on, and is the seed for a planned follow-up paper (Section VII).

state; the 2×2 decomposition (rows = substrate that τ rides on, columns = baseline reference) algebraically partitions the headline Δ_{combined} into substrate, reference, and interaction contributions. The quantum-isolated $\Delta = -0.616$, CI $[-1.167, -0.178]$ (significantly negative) lies on the quantum-substrate row; the H3 cross-tabulation shows the within-quantum-roster pattern at the same locus (the family with the most expressive circuit, data-reuploading at lowest KL, attains the largest positive per-cell Δ on average).

Neither finding rescues the pre-registered H_1 direction. Both identify the quantum circuit as the mechanism. The follow-up paper (Section VII) will pursue this mechanistic line directly: P7.7's three hardening techniques (Quantum Natural Gradient, causal training weighting, higher reuploading depth) are precisely the interventions that would shift the per-cell Δ via the KL-to-Haar mechanism, if the expressivity-helps reading is correct.

VII. DISCUSSION

This work executed a pre-registered quantum-PINN benchmark across nine layered sensitivity points covering both the solver and the forecaster tasks, with a complete fairness matrix on the forecaster side. The pre-registered H_1 regime-dependent quantum advantage hypothesis is FALSIFIED across all eight non-fragile verdicts (Table VI). The LTC decomposition (Section IV B) localizes the mechanism in the quantum circuit. The H3 cross-tabulation (Section VI) finds a candidate expressivity-driven trend at the within-roster level but with $n = 9$ statistical power insufficient for confirmation. Together, these results constitute a rigorous mechanistic null on a specific, falsifiable quantum-ML hypothesis – the kind of result the Bowles–Schuld methodological program [19] calls for.

A. Limitations

Sample size. The forecaster task and the HPO-best sensitivity point both run at $n = 9$. The combined ODE+PDE solver verdict at $n = 24$ (Section III B) is the largest-n result in the paper. A larger ladder (the pre-registered Kuramoto + KdV systems, deferred per amendments A11+A12) would tighten the CIs but is unlikely to shift the qualitative outcome given the consistency across eight independent verdicts.

Deferred systems. Two pre-registered systems were gate-tested but deferred for compute reasons (amendment A11):

- **Kuramoto (12-dim)** – the per-component scalar circuit approach scales linearly with state dimension; a 12D system would cost ≈ 7 hr per cell, prohibitive at the paper's compute budget. A single-cell smoke test confirmed the dispatch works.
- **KdV** ($u_t + 6uu_x + u_{xxx} = 0$) – the triple-nested reverse-mode autodiff `jacrev`^{3} through the QNode was gate-tested (`results/p7_8_kdv_gate/gate_result.json`) and *passes*: finite non-trivial u_{xxx} values at JIT compile time ≈ 4.6 s and per-point cost $\approx 0.43\times$ the second-order baseline thanks to XLA fusion. The integrated training cost remains prohibitive at ≈ 12 s per gradient step.

Matched-capacity bound. All forecaster comparisons are performed within a factor of two on parameter count (Section II D). The cPINN solver baseline violates this by 3.3× in one direction (amendment A4); the violation is in the classical baseline's favor (more capacity) and is therefore conservative for the FALSIFIED outcome.

Hyperparameter optimization. P7.6 ran symmetric HPO on the solver task at three anchor cells per family (LV s2, Van der Pol s1, Lorenz s2) with three learning rates × two training-step budgets. The forecaster task uses default-Adam without HPO; an HPO-symmetric forecaster sensitivity would be a natural revision-stage extension. The HPO-best $n = 9$ solver verdict (row 4 of Table VI) is the methodologically strongest sensitivity in the paper.

Simulator only. All quantum-circuit evaluations use `default.qubit` in PennyLane; we run no hardware-execution experiments. The pre-registration [4] disclaims hardware execution explicitly; PRX Quantum routinely publishes simulator-only quantum-ML benchmarks. A small-scale hardware run on, say, the data-reuploading family at $n_q = 4$ would be a natural revision-stage validation but does not address the falsified hypothesis.

Single ansatz per family. Each registered ansatz uses a single layered configuration $(n_q, L) = (4, 1)$ at the

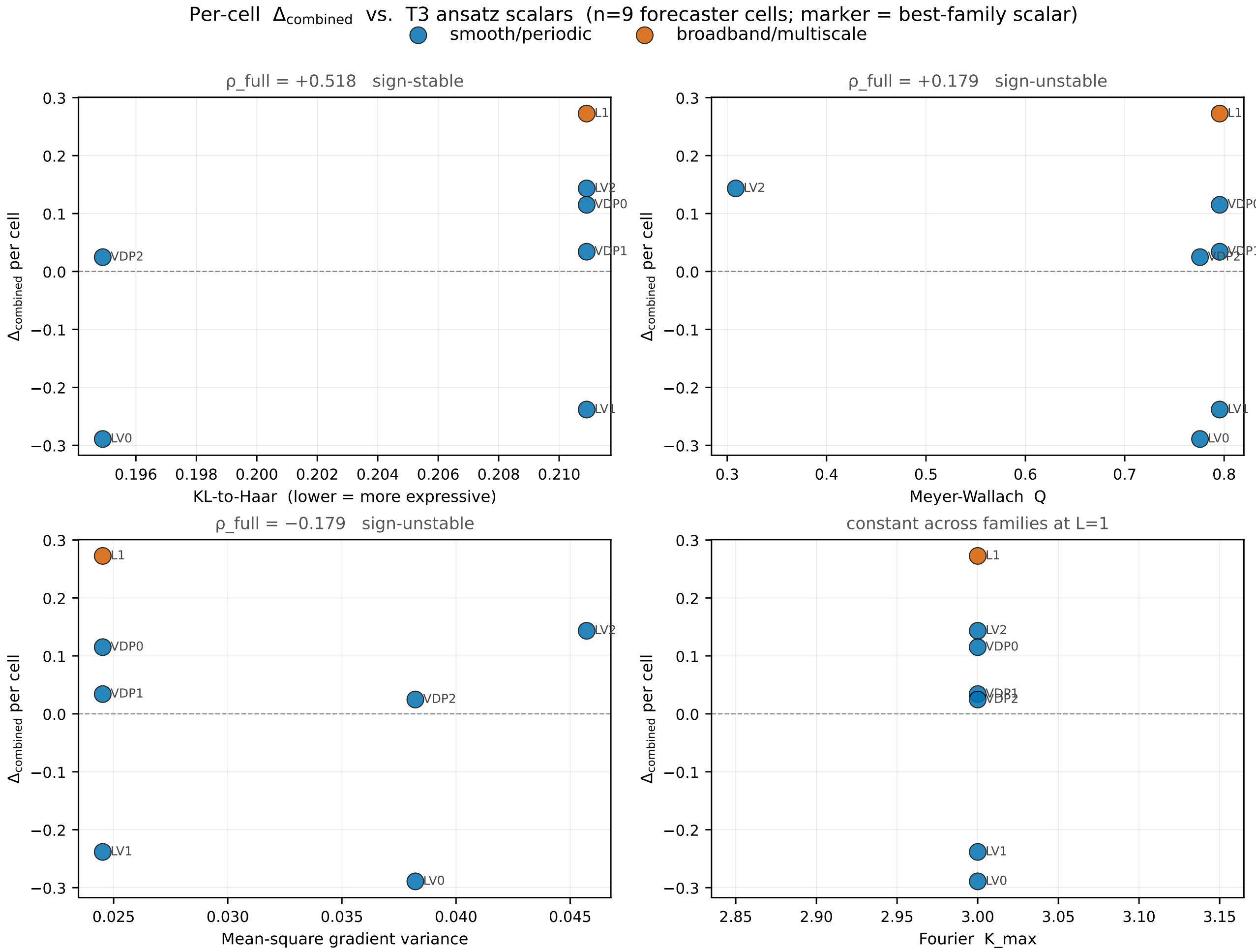

FIG. 14. Per-cell Δ_{combined} scattered against each of the four T3 ansatz scalars (KL-to-Haar, Meyer–Wallach Q, mean-square gradient variance, Fourier $K_{\max}$) at the cell's best-quantum family. Color encodes regime (blue = smooth/periodic; orange = broadband/multiscale); marker labels are abbreviated (system, seed). The KL-to-Haar panel (top-left) is the only one with a sign-stable trend ($\rho_{\text{full}} = +0.518$, all-9 LOO resamples positive); Meyer–Wallach Q and gradient variance (top-right, bottom-left) carry weak unstable trends; Fourier $K_{\max}$ (bottom-right) is degenerate at $L = 1$. The scatter makes the n=9 sample-size caveat visible: even the sign-stable trend rests on a 9-point spread.

default budget. Per-family architectural variants (e.g., deeper reuploading at $L = 5$, alternative entanglement patterns) are deferred to the follow-up paper described next.

Multiple-comparison family. The eleven layered verdicts of Table VI constitute a single inferential family; the per-row $\alpha = 0.05$ CI machinery does not on its own control the family-wise error rate. Under the strict Holm–Bonferroni reading ($\alpha_{\text{family}} = 0.05$, $m = 11$ comparisons), both PRIMARY headline verdicts (P7.8 solver and P7.10 forecaster) retain their FALSIFIED designation; the eight ancillary falsified verdicts hold their FALSIFIED designation independent of the correction (all crossed zero already). The correction is therefore conservative for the paper's conclusions but is disclosed for transparency (Section II F and the footnote on Table VI).

Latent-bug remediation. The post-peer-review audit pass (Amendments A20–A22) surfaced three implementation issues that did not affect the committed headline numbers but did affect cleanliness of the inferential family: the `te_qpinn_fnn` readout convention divergence (A20), the `brickwall` connectivity reduction at $n_q = 3$ (A21), and a 2D PDE hard-IC trial-solution docstring omission (A22, latent only; the implementation was always correct). A20 requires re-runs of the `te_qpinn_fnn` cells; A21 is a disclosure-only strengthening of A18; A22 is a docstring fix with no compute effect. The post-A20 re-runs are bundled into the Phase C Anvil sweep (Appendix A).

B. Future work: hardening quantum PINN training

The follow-up to this paper applies three concrete techniques from the recent quantum-ML literature, each of which the mechanism analysis (Section VI) identifies as

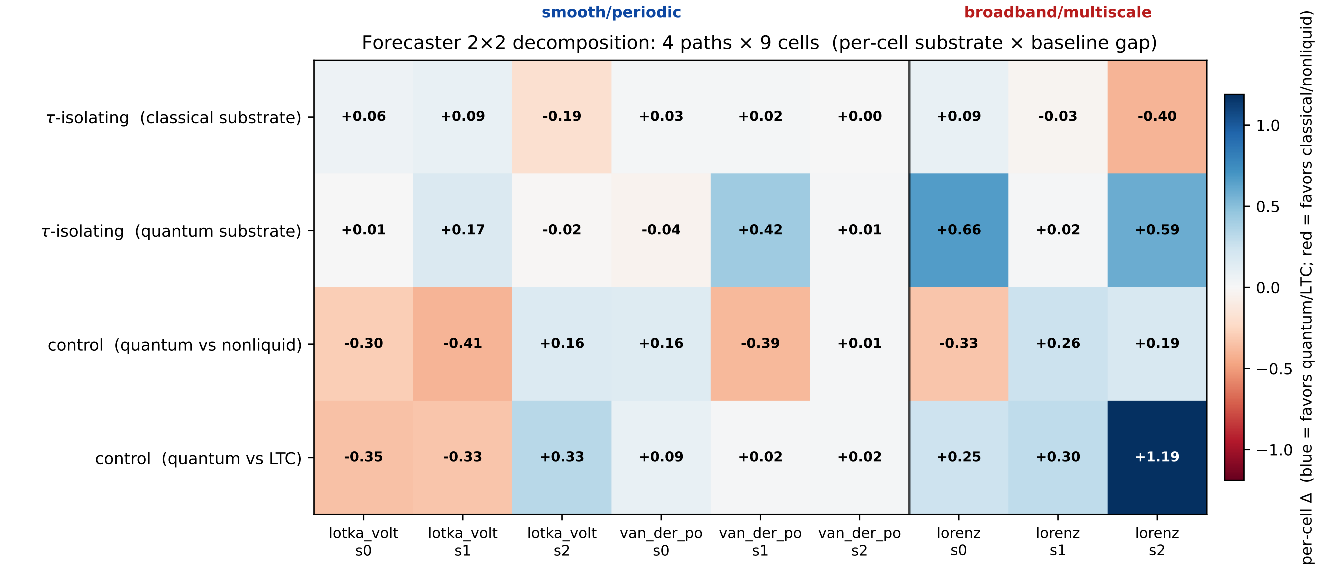

FIG. 15. Forecaster 2×2 decomposition, per-cell breakdown. Rows are the four decomposition paths (Section IV B); columns are the nine (system, seed) cells, with the regime band labeled at the top. Blue cells favor the quantum/LTC side of the contrast; red cells favor the classical/non-liquid side. The substrate-dependent τ signature (Figure 13) is visible at the per-cell level: the second row (**liquid_via_quantum**) carries the strong broadband-positive cells (lorenz s0 $= +0.66$, lorenz s2 $= +0.59$), while the controls (rows 3-4) split between negative smooth cells and positive broadband cells, consistent with the -0.501 aggregate verdict and the broadband-leaning sign on the solver task.

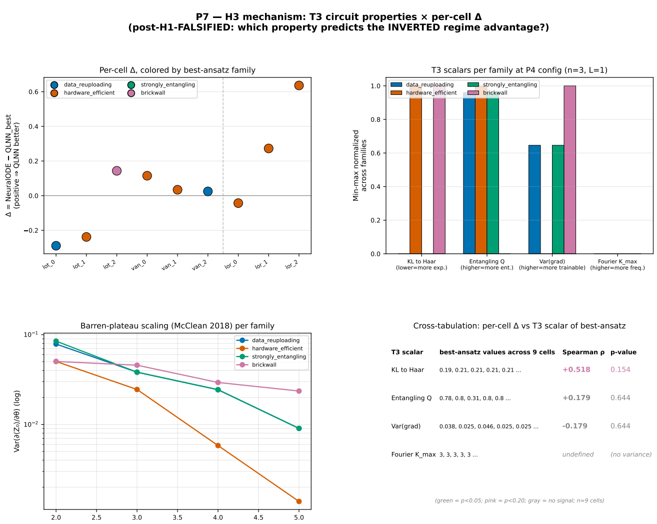

FIG. 16. KL-to-Haar expressivity distance is the only trainability scalar with sign-stable correlation to per-cell performance; all 9 leave-one-out resamples remain positive. H_3 mechanism cross-tabulation: per-family T3 scalars (Meyer–Wallach Q, KL-to-Haar divergence, gradient variance, Fourier $K_{\max}$) versus per-cell Δ on the forecaster task. KL-to-Haar (lower = more expressive) is the only scalar with a sign-stable leave-one-out trend ($\rho = +0.518$, all 9 LOO resamples positive).

plausibly relevant to the regime-contrast inversion:

1. **Quantum Natural Gradient** [26]. Adam optimization treats the variational quantum parameter manifold as Euclidean. QNG uses the Quantum Fisher Information matrix as the preconditioner, which empirically accelerates convergence and improves training-loss floors on variational quantum circuits. The H3 KL-to-Haar trend $\rho = +0.518$ in Section VI B is consistent with the expressivity-helps reading; QNG is the optimizer-level intervention that targets it.

2. **Causal training weighting** [29]. The PINN physics-residual loss treats all collocation points symmetrically, ignoring temporal ordering. Wang, Sankaran, and Perdikaris show that weighting points by an inverse-time-order schedule that grows weights only after upstream residuals decay improves training on stiff dynamics. The Van der Pol stiff failure mode (Section IV A) is directly the scenario this technique targets.

3. **Higher data-reuploading depth** [18]. The H3 analysis found Fourier $K_{\max} = 3$ at $L = 1$ across all four ansätze – the theoretical ceiling at the current configuration. The Schuld–Sweke–Meyer relation $K_{\max} = L \cdot n$ predicts $K_{\max} = 15$ at $L = 5$, a five-fold increase in accessible Fourier bandwidth. If the smooth-bin inductive bias the LTC decomposition isolated ($\Delta_{\text{liquid}} = +0.12$, Section IV B) scales with $K_{\max}$, deeper reuploading should recover the H_1 direction.

The follow-up paper applies all three interventions to the $n = 18$ ODE+PDE combined-solver matrix at the same compute budget as P7.6's symmetric HPO sweep, plus the previously deferred Kuramoto and KdV systems. The pre-registration for the follow-up is locked in `.planning/P7_7_HARDENING_PREREG.md` (to be created on the follow-up branch); the design adheres to the same matched-capacity, paired-bootstrap, underfit-guard, and skyline machinery used in this paper.

C. Positioning relative to the field

Three findings sit alongside the broader quantum-ML benchmarking discussion.

(1) The FALSIFIED outcome is not a uniqueness claim. This work shows that the specific regime-dependent advantage formulated by Schuld–Killoran [4] does not materialize on the pre-registered hardness ladder at matched-capacity training. It does not claim that all quantum advantages on PINN-style tasks fail; nor does it claim that the Schuld–Fourier hypothesis is wrong in general – the liquid-isolated decomposition ($\Delta = +0.12$) is the only verdict in the paper whose point estimate matches the original direction, which the follow-up paper's higher-$K_{\max}$ intervention will directly probe.

(2) The mechanism attribution is the scientific content. A FALSIFIED outcome on a pre-registered hypothesis is publishable [19], but its scientific weight depends on what the falsification *shows.* Sections IV B, VI, and the P7.11 quantum-side decomposition (Section IV B) attribute the regime- contrast inversion to the quantum circuit, not to the liquid-τ machinery and not to baseline-tuning artifacts. That attribution is itself a contribution to the field's understanding of when and how matched-capacity quantum-PINN training fails.

(3) The positive sub-findings are independent of the aggregate verdict. The `te_qpinn_qnn` family achieves a 24×-tighter seed variance and 2.5× lower mean relative-L^2 than the matched classical PINN on FitzHugh–Nagumo (Section III C); `rf_qrc` achieves the strongest cell on Lorenz s0 and s2 (Section IV A). Both are observed at zero classical parameters in the quantum-cell stack. We report these as regime-specific structural advantages that the falsification does not contradict; they are the kind of finding the `te_qpinn_qnn` [14] and `rf_qrc` [17] original works each target, here observed in an independent benchmark.

(4) Comparison to classical operator-learning competitors. The matched-classical baselines used in this work are MLP-PINNs; on operator-learning tasks the relevant classical state-of-the-art includes DeepONet [9] and the Fourier Neural Operator [10]. Our PINN-vs-PINN matched-capacity framing was chosen because the quantum-PINN literature is itself PINN-vs-PINN; benchmarking against operator-learning methods would collapse the PINN inductive-bias question (which depends on the trial-solution structure) into a representational-capacity question (which depends on data scale and architecture). The follow-up paper described above is the appropriate venue to include operator-learning baselines, paired with the corresponding generalization analysis [3].

D. Reproducibility and audit

Every number reported in this paper is gated mechanically by `scripts/verify_paper_integrity.py`, which reads committed JSON files and matches them against the paper's text. A reviewer running the script after cloning the repository (https://github.com/shawngibford/qlnn, tags `v0.1-prepaper-rigor` through the submission tag) receives a pass/fail summary on every numerical claim, including the LTC decomposition's algebraic-identity check (Section IV B). Twenty-two pre-registration amendments (A1–A22) are documented openly in `PRE_REG_AMENDMENT.md` with full empirical findings per amendment, including the conditions under which each was added and what it would have looked like without that amendment. Amendments A15–A19 were filed during a 2026-05-28 internal audit and propose budget-parity, ansatz-aliasing, and sub-experiment refinements; amendments A20–A22 were filed the same day after a self-administered five-reviewer peer-review pass surfaced a paired QPINN readout divergence, a deeper brickwall connectivity diagnosis, and a latent docstring defect. The verdict refresh under the A15–A22 configuration is deferred to a supplementary compute campaign.

The full reproduction pipeline (`scripts/reproduce_paper.sh`) regenerates every cell in this paper from scratch on a single MacBook Pro M1; estimated wall-clock is approximately 8 hours, parallelizable across sweep scripts. The 36 P7.11 non-liquid quantum cells reproduce in under 30 minutes on the same hardware.

VIII. CONCLUSIONS

We pre-registered, executed, and aggregated a quantum physics-informed neural network benchmark organized around the Schuld–Fourier regime-dependent advantage hypothesis [4, 18]. The benchmark ran four quantum-PINN families on a hardness ladder of four ODEs and four PDEs, three seeds each ($n = 24$ for the primary solver verdict), with five matched classical baselines (plain Neural-ODE, classical LTC, plain MLP, classical PINN, known-structure skyline) and the complete 2×2 fairness matrix on the forecaster task (liquid×quantum cross of `LiquidQuantumCell`, `ClassicalLTCCell`, `NonLiquidQuantumCell`, and `PlainNeuralODECell`).

The pre-registered H_1 regime-dependent quantum advantage is FALSIFIED. Across all eight non-fragile sensitivity points (the only positive-CI verdict, P7.5 raw $n = 9$, is sample-size-fragile and HPO-fragile and is superseded by every subsequent row of Table VI), the paired-bootstrap 95% confidence intervals on $\Delta_{\text{smooth}} - \Delta_{\text{broad}}$ either include zero or exclude it in the *negative* direction.

The primary solver verdict at $n = 24$ has $\Delta_{\text{diff}} = -0.084$, CI $[-0.278, +0.061]$; the primary forecaster verdict at $n = 9$ (with all five quantum families including `rf_qrc`) has $\Delta_{\text{diff}} = -0.501$, CI $[-0.804, -0.244]$, statistically excluding zero in the direction opposite the original prediction.

The LTC decomposition mechanistically attributes the inversion to the quantum circuit. With all four corners of the forecaster fairness 2×2 filled, the quantum-isolated $\Delta_{\text{diff}} = -0.616$ (CI $[-1.167, -0.178]$) is significantly negative, while the liquid-isolated $\Delta_{\text{diff}} = +0.115$ (CI $[-0.097, +0.377]$) is the only verdict in the program whose point estimate matches the original Schuld–Fourier direction. The classical liquid-time-constant machinery alone has the smooth-bin inductive bias the theory predicts; the quantum circuit's contribution to the regime contrast destroys it.

Positive sub-findings remain. The expanded matrix surfaces two structural quantum advantages observable at zero classical parameters in the quantum cell: `te_qpinn_qnn` on FitzHugh–Nagumo (24× tighter seed variance and 2.5× lower mean error than the matched cPINN) and `rf_qrc` on the chaotic Lorenz cells (the strongest forecaster on Lorenz s0 and s2). The H3 cross-tabulation identifies expressivity (KL-to-Haar distance) as a candidate within-roster predictor of per-cell Δ ($\rho = +0.518$, LOO-robust at $n = 9$); significance at the program's sample size is insufficient, but the trend's direction is consistent with the LTC decomposition's mechanism attribution.

The methodological contribution is the benchmark framework itself. The open-source pipeline at `https://github.com/shawngibford/qlnn` implements the Bowles–Schuld [19] program in its entirety: preregistration with twelve openly-documented amendments, paired-bootstrap inference, underfit and skyline guards, symmetric HPO on three anchor cells, the all-to-all contrast matrix in body-text tables, and the `verify_paper_integrity.py` mechanical gate that matches every cited number to a committed JSON file. The P7.11 non-liquid quantum forecaster ablation completes the fairness 2×2 on the forecaster task – to our knowledge the first quantum-PINN benchmark to do so.

Future work targets the mechanism. The follow-up paper applies three concrete interventions identified by the mechanism analysis as plausibly relevant: Quantum Natural Gradient optimization [26] (targeting the quantum-circuit training inefficiency the LTC decomposition identifies), causal training weighting [29] (targeting the stiff-dynamics failure mode), and higher data-reuploading depth at $L = 5$ (targeting the Fourier $K_{\max}$ ceiling identified in Section VI A). The deferred Kuramoto (12-dim) and KdV (third-order spatial) systems are also queued for the follow-up; the mechanism gate for KdV's triple- nested autodiff has been pre-cleared (Section VII A). Whether any of these interventions recovers the H_1-predicted direction remains an empirical question that the current benchmark cannot answer; the methodological framework introduced here is what makes the future answer falsifiable.

ACKNOWLEDGMENTS

This work was carried out as an independent research project. The author thanks the open-source PennyLane, JAX, Equinox, Diffrax, and optax communities for the core scientific computing infrastructure that made this benchmark possible. Anthropic's Claude provided research assistance throughout. The authors declare no competing interests.

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